

**Software Requirements
Specification**

IEEE 830 / ISO/IEC/IEEE 29148 Format

**BIOSPHERE: AN OFFLINE-FIRST BIOLOGICAL MUSEUM MANAGEMENT
SYSTEM WITH QR-BASED EXHIBITS AND AUGMENTED REALITY**

Version 0.10

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Revision History

Date	Version	Description	Author
July 25, 2026	0.9	Reformatted the current BioSphere SRS into the required IEEE 830 / ISO/IEC/IEEE 29148 structure; consolidated confirmed requirements, corrected the public Gallery and approved FAQ-assistant scope, added specimen-lot grouping, clarified restricted developer functions, and retained unresolved items for validation.	BioSphere Project Team
July 27, 2026	0.10	Simplified the MVP by removing SMS integration, limiting offline functionality to cached specimen viewing and new Uncataloged text-only draft capture, defining email as outbound-only without external mailbox synchronization, and making the Home, Gallery, Visit, and About pages static frontend content. Updated the database design by separating specimen quantity, condition, and location into the SpecimenLot table and adding SpecimenLotTransaction for lot movements, condition changes, splits, merges, and quantity adjustments.	BioSphere Project Team

August 11, 2026	0.11	Clarified the curator-approved selection guidelines for specimens considered for AR presentation; established that AR selection remains under curator discretion and is not determined through an automated scoring or ranking mechanism.	BioSphere Project Team
Pending approval	1.0	Baseline release after curator, adviser, privacy, deployment, and report.	BioSphere Project Team

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for BioSphere version 0.9, a web-based biological museum management system being developed for the University of San Carlos Biological Museum (USC-BM). It serves as the baseline for system design, implementation, testing, client validation, pilot deployment preparation, turnover, and acceptance.

BioSphere consists of a public museum website, an authenticated curator Progressive Web Application (PWA), a restricted developer interface, a shared backend and database, controlled public QR exhibit pages, and selected browser-based augmented reality experiences. The system centralizes specimen documentation, storage-location tracking, visitor communications, reporting, audit logging, backup and recovery, and selected offline inventory operations.

BioSphere will not replace the University of San Carlos campus-entry or ISMIS process, will not provide public access to the complete internal specimen inventory, and will not provide unrestricted or general-purpose chatbot responses.

1.2 Document Conventions

- The word "shall" indicates a mandatory requirement. "Should" indicates a recommended requirement, and "may" indicates an optional or future capability.
- Each functional requirement is identified using REQ-<section>-<number>, such as REQ-4.4-01.
- Priority is labeled High, Medium, or Low. High requirements are part of the MVP baseline; Medium requirements are important but may be deferred only through approved scope control; Low requirements are future enhancements.

- Terms such as curator-approved, university-approved, and formally authorized require documented approval by the appropriate USC-BM or USC representative.
- Items explicitly marked subject to validation remain provisional and must be confirmed before Version 1.0.

1.3 Intended Audience and Reading Suggestions

Audience	Recommended Reading
USC-BM curator and client representative	Sections 1, 2, 4, 5.5, 6, and Appendix B for scope, workflows, responsibilities, rules, and acceptance.
Faculty adviser and capstone panel	Sections 1-6 and Appendices A-B for feasibility, technical depth, limitations, quality requirements, and defensibility.
Development team and software architect	Sections 2-6 and Appendix B for architecture, interfaces, features, data relationships, constraints, and deployment.
Testers and evaluators	Sections 4-6 and Appendix B.4 for functional test derivation, non-functional targets, and traceability.
Future USC/DCISM maintainers	Sections 2.4-2.6, 3.2-3.4, 5, 6.2, and turnover requirements.
Public-facing content reviewers	Sections 3.1, 4.10-4.13, 5.3, and 5.5 for publication, privacy, FAQ, QR, and AR boundaries.

1.4 Product Scope

BioSphere is a new standalone system for USC-BM. Its primary purpose is to replace fragmented paper-based and spreadsheet-based specimen documentation with a centralized, searchable, backed-up, and auditable

inventory while preserving the curator's approval authority over specimen data and public content.

The MVP includes:

- Authentication, separate curator accounts, secure curator onboarding through an email invitation and password-setup process, account recovery, curator succession, and restricted developer access.
- Specimen cataloging, images, taxonomy and descriptive data, Uncataloged/Cataloged/Archived states, spreadsheet or CSV import, and possible duplicate warnings.
- Specimen-lot grouping for quantity, condition, and location changes without losing total-quantity integrity.
- Hierarchical storage locations using rooms or galleries, cabinets, drawers or shelves, boxes, vials, and other curator-approved units.
- Search, filtering, sorting, five core report types, DOCX/PDF/CSV/print exports where applicable, and audit history.
- General Inquiry and Visit Request workflows with multiple preferred schedules, in-system curator alerts, email notifications to the authorized curator when new submissions are received, and outbound email replies or status updates initiated by the curator through BioSphere. Incoming replies received through an external email account are not imported into BioSphere.
- A static public website with Home, Gallery, Visit, and About areas; no Events page, no searchable public inventory, and no curator-facing content-management system for these pages.
- A controlled knowledge-based FAQ assistant that answers only from curator-approved knowledge and falls back to the General Inquiry form.
- Unlisted QR exhibit pages and optional WebAR for selected approved exhibits with normal-page fallback.
- Limited offline inventory support for viewing and searching previously synchronized specimen records and creating new Uncataloged text-only drafts for synchronization after reconnection.

- Audit logging, configurable backup and recovery, documented restoration testing, deployment documentation, and turnover support.

The MVP excludes a student OJT module, direct ISMIS integration or automatic campus-entry approval, native mobile applications, public access to the complete inventory, AR for every specimen, support for other USC museum units, unrestricted chatbot behavior, a general-purpose administrator dashboard, and indefinite maintenance by the student team after turnover.

1.5 References

Ref#	Document Title	Version/Date	Source
[1]	ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering.	2018	International Organization for Standardization.
[2]	CAPSTONE_SRS-3.docx - Software Requirements Specification IEEE 830 / ISO/IEC/IEEE 29148 Format.	Provided July 2026	Required USC capstone SRS template.
[3]	SRS CURRENT 1 25 AM JULY 25 2026 - BioSphere Software Requirements Specification.	Version 0.9, July 25, 2026	BioSphere project team working draft.
[4]	Biosphere: An Offline-First Biological Museum Management System With Qr-Based Exhibits And Augmented Reality.	July 2026	BioSphere capstone manuscript and approved project scope.
[5]	USC Biological Museum curator interview and workflow notes.	July 8, 2026	Internal project records; client

			representative: Fidel Bendanillo.
[6]	Republic Act No. 10173 - Data Privacy Act of 2012.	2012	National Privacy Commission / Republic of the Philippines.
[7]	Implementing Rules and Regulations of Republic Act No. 10173.	2016	National Privacy Commission.
[8]	BioSphere UI/UX mockups and HTML prototypes.	July 2026	Internal project design artifacts.

2. Overall Description

2.1 Product Perspective

BioSphere is a new system rather than an extension of an existing museum application. It will consolidate the USC-BM curator's paper and spreadsheet workflows into a shared database and controlled media storage. It will coexist with, but will not replace, the USC campus-entry and ISMIS workflow. The curator will manually use the approved visit-request information when completing the external university process.

The solution uses two separate Next.js and TypeScript frontends: a public website and an authenticated curator PWA. Both communicate with a shared NestJS TypeScript modular-monolith backend through a documented REST API. PostgreSQL stores relational data through Prisma, while approved images, documents, QR assets, and 3D models are stored in controlled object storage. The curator PWA uses IndexedDB through Dexie and a service worker to cache selected specimen records and store newly created Uncataloged text-only drafts until they can be submitted after reconnection.

A minimal restricted developer interface may be provided for the two approved technical responsibilities: authorized curator-access administration and AR-asset deployment. It shall not become a general-purpose administration dashboard and shall not expose routine curator operations.

2.2 Product Functions

- Authenticate curators and restricted developers; enforce role boundaries and secure sessions.
- Onboard authorized curators through secure email invitations, maintain separate credentials, support password change and email-based account recovery, activate or deactivate accounts, and support curator handover.
- Create, validate, import, search, edit, archive, and export specimen records and images.
- Group quantities into specimen lots and preserve quantity consistency during partial moves or condition changes.
- Manage a self-referencing hierarchy of rooms/galleries, cabinets, drawers/shelves, boxes, vials, and other storage units.
- Display dashboard summaries, pending work, alerts, connection status, and the synchronization status of offline-created drafts.
- Alert the authorized curator about new submissions and send curator-initiated outbound replies or status updates to visitors by email. BioSphere shall not retrieve or synchronize incoming messages from an external email mailbox.
- Display approved static website and Gallery content; allow authorized curator management of FAQ knowledge and QR exhibit pages; and provide selected WebAR experiences.
- Maintain curator-approved FAQ knowledge and provide controlled rule-based answers with General Inquiry fallback.
- Generate the five approved reports and supported editable or exportable outputs.
- Record auditable activity, execute backup and recovery procedures, and support deployment and turnover.

2.3 User Classes and Characteristics

User Class	Characteristics	Privileges/Access
Museum Curator	Primary authenticated operational user; may be non-technical; uses the system regularly for specimen, location, visitor, FAQ-knowledge, QR-exhibit, reporting, and handover tasks	Full curator operations within approved scope; separate credentials; cannot alter protected audit entries or developer-only deployment functions.
Public Visitor	Unauthenticated visitor using desktop or mobile browsers; no training expected.	View approved website, Gallery, QR, and AR content; use FAQ assistant; submit inquiry or visit request; no internal inventory access.
Developer	Restricted technical actor with software and deployment knowledge; infrequent use.	Initial curator onboarding; formally authorized curator-access disabling/deactivation; approved AR asset upload/update/activation; no routine museum content or inventory management.

2.4 Operating Environment

Component	Operating Environment
Curator devices	Desktop or laptop computers using a current supported version of Google Chrome, Microsoft Edge, or Mozilla Firefox. Responsive tablet or mobile access may be used

	for selected tasks. Browser storage must be enabled to support Progressive Web Application caching and offline functions.
Public devices	Public exhibit pages may be accessed through supported desktop, tablet, or mobile browsers. Visitors may use their device's available QR-reading capability to open an exhibit page, but QR scanning will not be the only access method because each approved exhibit label will also provide a human-readable direct URL. Browser-based WebAR will require a compatible device, supported browser capabilities, and camera permission. If these requirements are not met, visitors will still be able to access the standard exhibit page and its approved non-AR content.
Server	A university-approved Linux server, or the Department of Computer, Information Sciences, and Mathematics Server (DCISM) or approved cloud-hosting environment capable of running the application services, PostgreSQL database connectivity, Docker Compose, and Nginx. Final deployment is subject to DCISM infrastructure assessment and approval
Frontend	Two separate Next.js and TypeScript applications: the public website and the curator Progressive Web Application. Tailwind CSS and shadcn/ui will support interface development.
Backend	A NestJS and TypeScript modular monolith that exposes documented REST API endpoints, with API documentation generated using the OpenAPI specification.

Database and ORM	PostgreSQL managed through Prisma ORM. Deployment may use an approved Supabase PostgreSQL arrangement or another USC-approved PostgreSQL environment.
Object storage	Supabase Storage or another approved S3-compatible object-storage service for specimen images and approved three-dimensional or WebAR assets.
Offline components	IndexedDB managed through Dexie for cached specimen records and offline-created Uncataloged drafts, together with Workbox or an equivalent service worker for application-resource caching. Existing specimen records will remain read-only while offline.
Deployment tooling	Git and GitHub for version control, GitHub Actions for approved automated workflows, Docker and Docker Compose for application packaging, and Nginx as the reverse proxy. Final continuous integration and deployment procedures and server permissions will depend on USC approval.

2.5 Design and Implementation Constraints

- The implementation shall use the confirmed TypeScript-based architecture: separate Next.js public and curator applications, a NestJS modular-monolith backend, PostgreSQL, Prisma, IndexedDB/Dexie, and a service worker/Workbox approach.
- The system shall be web-based and responsive; a native Android or iOS application is outside the MVP.
- The public website shall use the approved navigation labels Home, Gallery, Visit, and About and shall not include an Events page or a public searchable inventory.

- Public QR exhibit pages shall remain unlisted from ordinary site navigation and shall display only curator-approved fields.
- AR shall use browser-based WebAR or the device model viewer through an approved library such as <model-viewer>; custom 3D-model production is not mandatory for every exhibit.
- Offline-first behavior shall be limited to viewing and searching previously synchronized specimen records and creating new Uncataloged text-only drafts. Editing existing records and all other system operations shall require connectivity.
- The solution shall comply with applicable Philippine privacy requirements and USC policies, subject to final approved privacy notice, retention, access, and deployment procedures.
- The student team shall plan for a six-month development and validation period and shall avoid microservices or other architecture that cannot be completed, tested, and explained within the capstone timeline.
- The final deployment, domain, TLS certificate, authorized email sender, storage quota, server resources, and backup destination require USC/DCISM authorization.

2.6 Assumptions and Dependencies

Type	Assumption or Dependency
Assumption	USC-BM will identify the authorized curators and designated client representative responsible for validating specimen records and approving public website content, FAQ knowledge, QR exhibit pages, and selected AR exhibits.
Assumption	USC-BM will provide approximately 500 curator-validated specimen records from the museum's main gallery for data preparation, system testing, and pilot implementation. The final

	number may vary slightly depending on record availability and validation results.
Assumption	Users will access BioSphere through current browsers that support the required Progressive Web Application, IndexedDB, service-worker, and browser-based WebAR functions. WebAR availability may vary depending on the device, browser, camera permission, and hardware capability.
Dependency	The official accession-number format, meaning, uniqueness rule, and assignment procedure remain subject to final curator validation.
Dependency	Data migration quality depends on the completeness, consistency, accuracy, and legibility of the museum's existing paper and spreadsheet records.
Dependency	General Inquiry and Visit Request emails, curator onboarding invitations, password-recovery emails, object storage, domain and TLS configuration, and backup services depend on approved third-party services or USC/DCISM infrastructure, permissions, and availability.
Dependency	Offline functionality depends on available browser storage, supported browser and device behavior, device health, and successful authenticated reconnection to the server. Offline-created drafts remain stored only on the device until synchronization succeeds.
Dependency	Visit-request processing remains dependent on the existing USC campus-entry procedure, which is completed outside BioSphere by authorized museum or university personnel.

Dependency	Selected AR exhibits depend on the availability of approved, legally usable, optimized, and technically compatible three-dimensional assets.
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3. External Interface Requirements

3.1 User Interfaces

The interfaces shall use a consistent USC-inspired white-and-green visual language, clear labels, responsive layouts, visible status feedback, confirmation prompts for high-impact actions, accessible contrast, and keyboard-usable controls where applicable. The public website and curator PWA are separate applications but shall retain consistent branding and terminology.

3.1.1 Public Website Interface

The public website shall contain Home, Gallery, Visit, and About pages, a visible Plan Your Visit action, a footer with approved contact and location information, and access to the FAQ assistant. It shall not contain an Events page or a searchable public inventory. The Gallery presents selected educational collection content rather than the museum's complete specimen records.



Figure 1. Public Gallery-Page Draft (formerly labeled Exhibits)

As shown in Figure 1, the Gallery page uses the same public navigation and hero banner pattern as the Landing Page, followed by alternating image-text sections for each Featured Area with supporting icon-labeled highlights, a "Why Visit Our Exhibits" benefits section, and a closing "See It All in Person" call-to-action banner leading to the standard footer.



Figure 2. Public About-Page Draft

As shown in Figure 2, the About page follows the same header/hero pattern as other public pages, then presents an "Our History" section paired with a "Through the Years" highlight, three Mission/Education/Conservation info cards, and a Curatorial Team section listing staff profiles with photo, role, contact details, and LinkedIn links, ending in the standard footer.

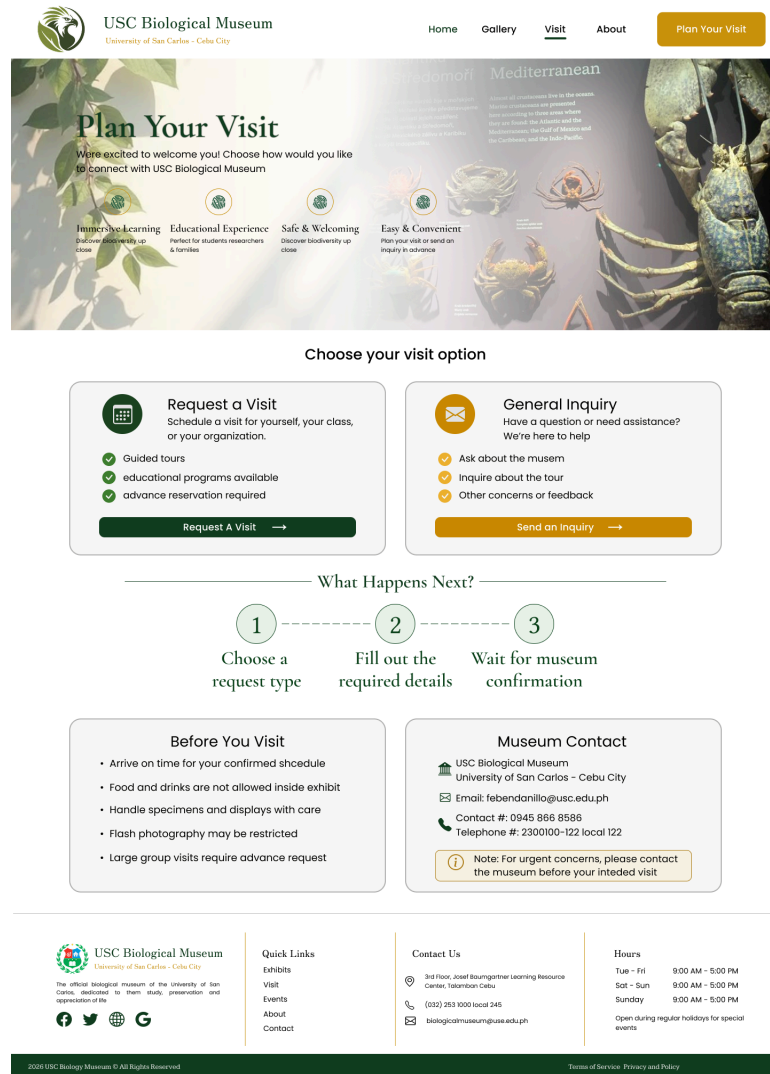


Figure 3. Public Visit-Page Draft

As shown in Figure 3, the Visit page presents two action cards, "Request a Visit" and "General Inquiry", each with feature checklists and a CTA button, followed by a 3-step "What Happens Next?" process guide, and closing informational panels for "Before You Visit" guidelines and "Museum Contact" details before the standard footer.

3.1.2 Public Inquiry and Visit-Request Interfaces

As shown below in Figures 5 and 6, the General Inquiry and Visit Request interfaces shall appear as dedicated pages or accessible modal forms, clearly identify required fields, validate entries before submission, present a privacy

notice and consent mechanism, and display a success or error state. Cancelling a modal shall not save the partially entered information unless a future approved draft behavior is implemented.

Request a Visit
Tell us about your planned visit to the museum

CONTACT DETAILS

Contact Person Name: Juan Dela Cruz
Email Address: name@school.edu.ph
Contact Number: 09XX XXX XXXX
Contact Number: University of San Carlos

Purpose of visit: Field trip, research, class requirement, etc.

SCHEDULE

Preferred date: mm/dd/yyyy
Start time: 9:00 AM
End time: 9:00 AM

VISITORS

Number of visitors: 4
Upload visitor list (if more than 5): name@school.edu.ph

VEHICLE

Yes, bringing a vehicle / No vehicle
Plate number: ABC 1234
Car brand: Toyota
Car type: Van, Sedan, School Bus

ADDITIONAL INFO

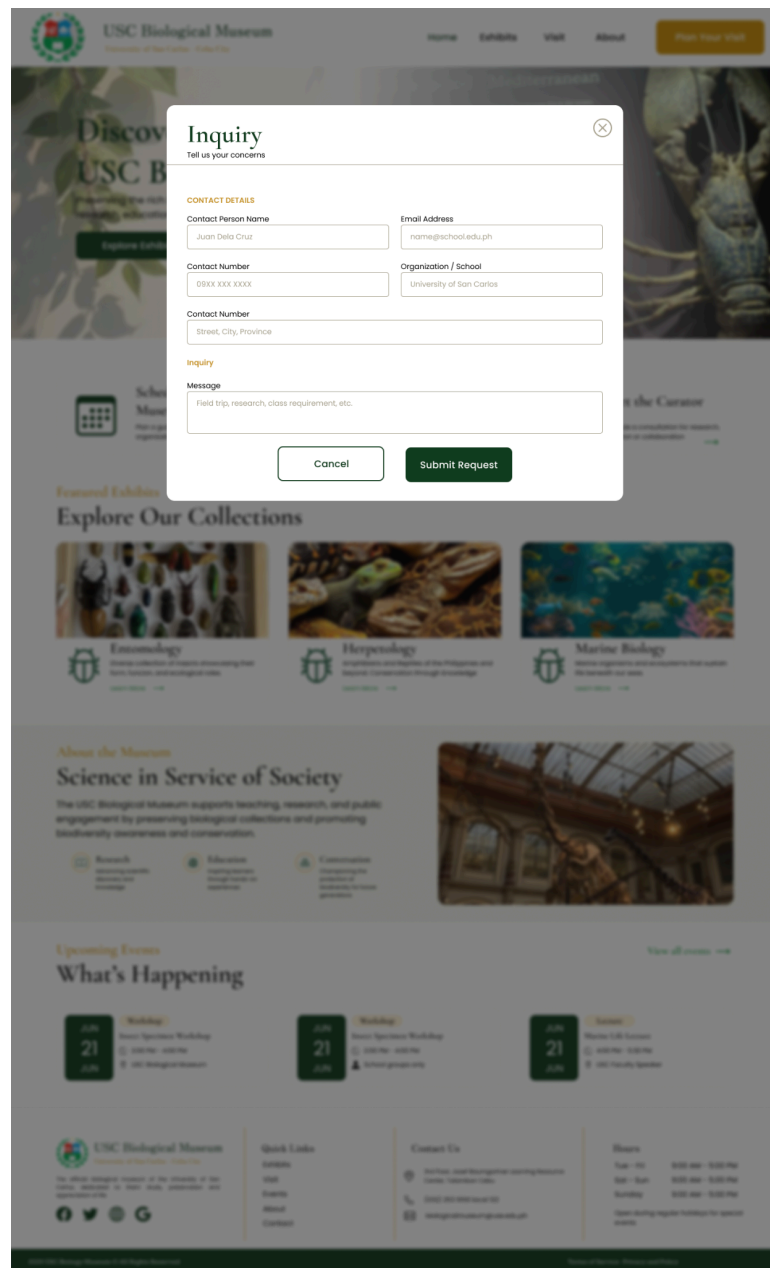
Equipment to bring: Cameras, notebooks, recording gear
Additional notes: Anything else we should know

Cancel Submit Request

Figure 4. Visit Request Form Draft.

As shown in Figure 4, the "Request a Visit" form on the public museum website is organized into sectioned fields Contact Details, Schedule, Visitors,

Vehicle, and Additional Info with dynamic visitor name fields, a toggle for vehicle details, and standard Cancel/Submit Request buttons at the bottom.



The image shows a screenshot of the USC Biological Museum website. A modal form titled "Inquiry" is open in the center. The form has a close button (X) in the top right corner. Below the title, it says "Tell us your concerns". The form is divided into two main sections: "CONTACT DETAILS" and "Inquiry".

CONTACT DETAILS

- Contact Person Name:** Juan Dela Cruz
- Email Address:** name@school.edu.ph
- Contact Number:** 0900 000 0000
- Organization / School:** University of San Carlos
- Contact Number:** Street, City, Province

Inquiry

Message: Field trip, research, class requirement, etc.

At the bottom of the form are two buttons: "Cancel" and "Submit Request".

The background of the website shows various sections: "Discover USC B", "Featured Exhibits", "Explore Our Collections" (with sub-sections: Entomology, Herpetology, Marine Biology), "About the Museum", "Science in Service of Society", "Upcoming Events", and "What's Happening".

Figure 5. General Inquiry Form Draft

As shown in Figure 5, the "Inquiry" modal on the public website presents a simplified form with Contact Details and an Inquiry Message field, paired with standard Cancel and Submit Request buttons.

3.1.3 Curator Inventory PWA Interface

The authenticated curator PWA shall provide a persistent navigation structure for Dashboard, Specimens, Cataloging, Locations, Visit Requests, General Inquiries, Exhibits/QR, FAQ Knowledge, Reports, Users, and Audit/Backup functions according to the final approved information architecture. The interface shall show the current user, connection/synchronization state, notifications, search access, and clear record status labels.

Inventory tables shall support search, filters, sorting, pagination or virtualized loading, and clear actions. Editing a specimen shall open the cataloging interface or a full record form rather than hiding complex specimen data inside a small modal. Location screens shall visualize the Room/Gallery - Cabinet - Drawer/Shelf - Box/Vial hierarchy and show specimens contained by valid leaf units.

The following figures present the current interface drafts and HTML prototypes used to guide implementation. Final interfaces may undergo minor usability refinements while remaining consistent with the approved functional requirements.

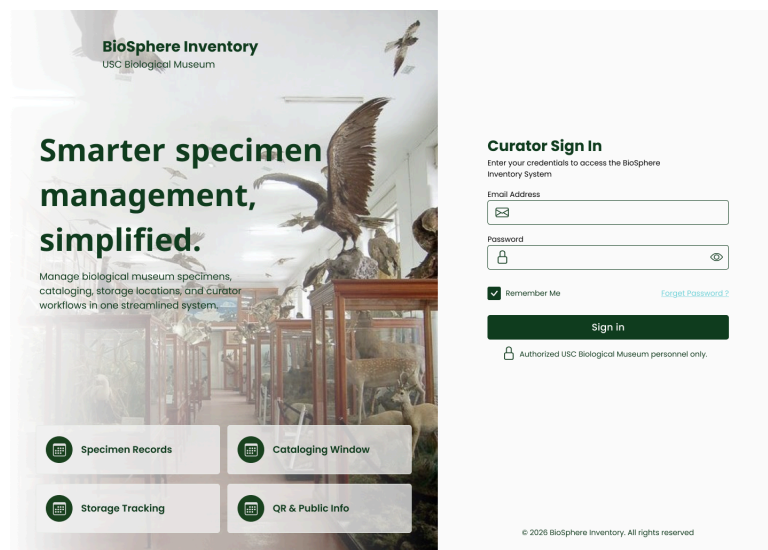


Figure 6. Login Page Draft

As shown in Figure 6, the Login screen uses a split-panel layout with a branded showcase image and feature highlights on the left and a Curator Sign In form on the right, containing Email Address and Password fields, a "Remember Me" checkbox, "Forgot Password?" link, and a "Sign in" button.



Figure 7. Dashboard Page Draft

As shown in Figure 7, the Dashboard presents summary metric cards (Total Specimens, Drafts, Needs Review, Duplicates, Pending Requests) followed by visual widgets a Specimen by Collection Type donut chart, a Cataloging Process line graph, and a Cataloging Queue Snapshot list alongside a Quick Actions panel (Add New Specimen, Continue Cataloging, Generate Report, QR Generator) for one-click access to core tasks. The lower section displays a Recent Specimen Records table, a Collection Health/Storage Overview with capacity bars and alerts, a Recent Cataloging Activity feed, and a Public QR Readiness summary, all consistently using the standard header (search bar, sync status, notifications, and user profile) found across every screen.

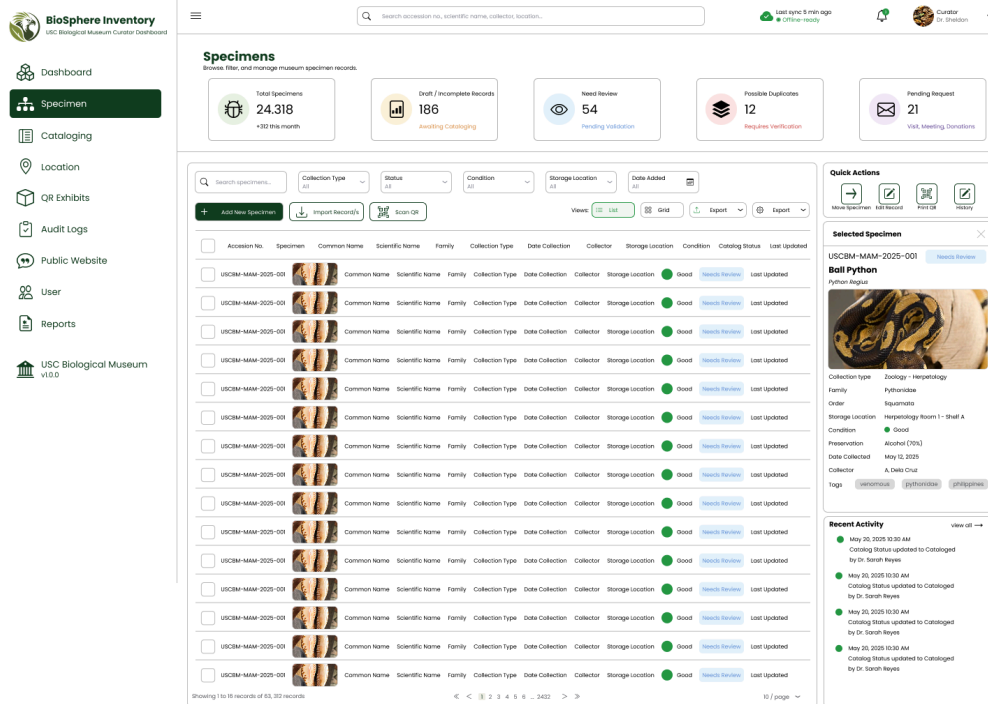


Figure 8. Specimen Page Draft

As shown in Figure 8, the Specimens page displays a summary dashboard of key metrics (Total Specimens, Drafts, Needs Review, Duplicates, Pending Requests) followed by a filterable, paginated data table listing specimen records with columns such as Accession No., Common Name, Scientific Name, Family, Collection Type, Collector, Storage Location, Condition, and Catalog Status. Standard controls include a global search bar, filter dropdowns (Collection Type, Status, Condition, Storage Location, Date Added), List/Grid view toggles, Import/Export/Scan QR buttons, and an "Add New Specimen" button, while a right-side panel shows Quick Actions, details of the currently selected specimen, and a Recent Activity log for audit tracking.

As shown in Figures 9, 10, 11, the "Add Specimen" modal presents a tabbed interface with Basic Info, Taxonomy, and Storage & Media sections, alongside standard header buttons (Save Draft, Generate QR, Mark as Cataloged, and Close). The Taxonomy tab captures classification fields including Kingdom, Phylum, Class, Order, Family, Genus, Species, Habitat, Conservation Status, Ecological Role, and Tag, while the Storage & Media tab includes a

Storage Unit dropdown, a Storage Overview field, and an option to add/upload specimen photos. All input fields follow a consistent labeled text-box layout for accessibility and ease of data entry across tabs.

The screenshot shows a web application interface for 'Morphology Inventory'. A modal window titled 'Add Specimen' is open, displaying the 'Basic Info' tab. The modal contains several input fields for specimen data: 'Accession No.' (with a dropdown), 'Collection Name', 'Common Name', 'Scientific Name', 'Specimen Category', 'Gender', 'Quantity', 'Condition', 'Collector', 'Donor', 'Collection Date', 'Collection Location', 'Preservation Method', 'Preservation Type', and 'Remarks'. Each field has a small 'Add' button next to it. The background shows a list of specimens with columns for accession number, image, and status.

Figure 9. Add Specimen - Basic Info Modal Draft

The screenshot shows the same 'Add Specimen' modal window, but with the 'Taxonomy' tab selected. This tab contains input fields for taxonomic information: 'Kingdom', 'Phylum', 'Class', 'Order', 'Family', 'Genus', 'Species', 'Conservation Status', 'Collector', 'Preservation Date', 'Tag', and 'Add Tag'. Each field has a small 'Add' button next to it. The background interface remains the same as in Figure 9.

Figure 10. Add Specimen - Taxonomy Info Modal Draft

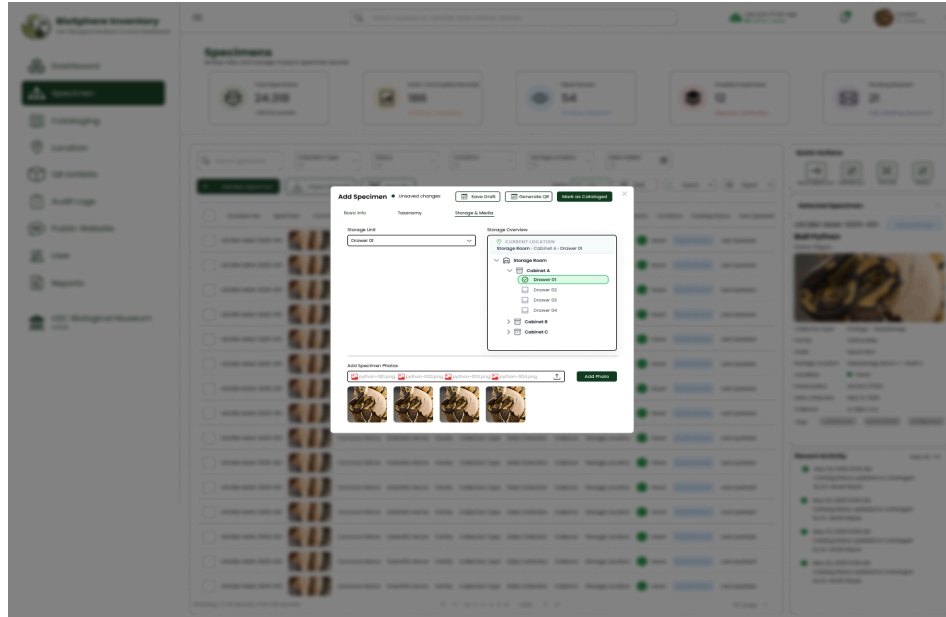


Figure 11. Add Specimen - Storage and Media Modal Draft

As shown in Figures 12, 13, 14, 15, 16, the Cataloging page allows curators to review, update, and validate existing specimen records after their initial creation in the Specimen workflow module. The page features a cataloging queue, editable record form, and validation panel with standard functions such as search, save, generate QR, mark as cataloged, and navigation controls, ensuring an efficient and accessible record maintenance workflow.

Accession No.	Specimen	Updated By	Last Updated	Changes	Status
USCBM-MAM-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 22, 2025 10:05 AM	Added habitat, logs	Cataloged
USCBM-MAM-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 22, 2025 10:05 AM	Added habitat, logs	Cataloged
USCBM-MAM-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 22, 2025 10:05 AM	Added habitat, logs	Cataloged
USCBM-MAM-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 22, 2025 10:05 AM	Added habitat, logs	Cataloged
USCBM-MAM-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 22, 2025 10:05 AM	Added habitat, logs	Cataloged

Figure 12. Catalogue Page - Basic Info Edit Draft

BioSphere Inventory
USC Biological Museum Curator Dashboard

Dashboard
Specimen
Cataloging
Location
QR Exhibits
Audit Logs
Public Website
User
Reports
USC Biological Museum v1.0.0

Welcome Back, Sheldon
Here's the current status of the collection, cataloging workflow, and storage today.

Pending Catalog 186
+ 12 since yesterday

Need Review 54
29.0% of total

Possible Duplicates 12
6.4% of total

Missing QR / Public Info 12
16.6% of total

Completed Today 21
+ 4 from yesterday

Cataloging Queue
View all →

Search queue

AS Draft Needs Review Missing Field

Possible Duplicate Ready for QR

USCIB-M-2025-1387
Papilio zalcion

USCIB-M-2025-090
Dasyprocta leprosa

USCIB-M-2025-0485
Dasyprocta leprosa

USCIB-MAM-2025-0287
Tamandua mexicana

USCIB-MAM-2025-0302
Cyrtodactylus

USCIB-M-2025-1387
Papilio zalcion

USCIB-M-2025-090
Dasyprocta leprosa

USCIB-M-2025-0485
Dasyprocta leprosa

Showing 1 to 6 records of 60 records

Catalog Entry • Unsaved changes Save Draft Generate QR Back to Cataloging

Basic Info Taxonomy Storage & Media Manage Specimen Storage

Origin: Argentina
Specimen: 123456
Date: 2025-05-23
Sex: Male
Species: Dasyprocta leprosa
Habitat: Terrestrial lowland forest
Ecological Role: Prey / Mesopredator
Tag: 123456

Validation & Preview

USCIB-MAM-2025-001
Bull Python (Python regalis)

Collection Type: Neotropical Storage Location: Room 1, Cabinet 4, Shelf A, Box 12 Condition: Good

Cataloging Checklist 4/4
Basic Information Completed
Taxonomy Completed
Storage Assigned
Preservation Recalculated

Duplicate Check
USCIB-MAM-2025-0287
USCIB-MAM-2025-0287
USCIB-MAM-2025-0287

QR / Exhibit Preview - Active
Bull Python
Python regalis
Habitat: Terrestrial lowland forest
Ecological Role: Prey / Mesopredator
Show more

Recent Activity
May 23, 2025 10:30 AM
Catalog entry created by Dr. Sarah Reyes
Basic info updated by Dr. Sarah Reyes

Recent Cataloging Activity
View all →

Accession No.	Specimen	Updated By	Last Updated	Changes	Status
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged

Showing 1 to 5 records of 60 records

Figure 13. Catalogue Page - Taxonomy Info Edit Draft

BioSphere Inventory
USC Biological Museum Curator Dashboard

Dashboard
Specimen
Cataloging
Location
QR Exhibits
Audit Logs
Public Website
User
Reports
USC Biological Museum v1.0.0

Welcome Back, Sheldon
Here's the current status of the collection, cataloging workflow, and storage today.

Pending Catalog 186
+ 12 since yesterday

Need Review 54
29.0% of total

Possible Duplicates 12
6.4% of total

Missing QR / Public Info 12
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Completed Today 21
+ 4 from yesterday

Cataloging Queue
View all →

Search queue

AS Draft Needs Review Missing Field

Possible Duplicate Ready for QR

USCIB-M-2025-1387
Papilio zalcion

USCIB-M-2025-090
Dasyprocta leprosa

USCIB-M-2025-0485
Dasyprocta leprosa

USCIB-MAM-2025-0287
Tamandua mexicana

USCIB-MAM-2025-0302
Cyrtodactylus

USCIB-M-2025-1387
Papilio zalcion

USCIB-M-2025-090
Dasyprocta leprosa

USCIB-M-2025-0485
Dasyprocta leprosa

Showing 1 to 6 records of 60 records

Catalog Entry • Unsaved changes Save Draft Generate QR Back to Cataloging

Basic Info Taxonomy Storage & Media Manage Specimen Storage

Storage Location: Room 1, Cabinet 4, Shelf A, Box 12
Storage Room: Cabinet 4, Drawer 12
Drawer 12
Drawer 13
Drawer 14
Cabinet 5
Cabinet 6

Validation & Preview

USCIB-MAM-2025-001
Bull Python (Python regalis)

Collection Type: Neotropical Storage Location: Room 1, Cabinet 4, Shelf A, Box 12 Condition: Good

Cataloging Checklist 4/4
Basic Information Completed
Taxonomy Completed
Storage Assigned
Preservation Recalculated

Duplicate Check
USCIB-MAM-2025-0287
USCIB-MAM-2025-0287
USCIB-MAM-2025-0287

QR / Exhibit Preview - Active
Bull Python
Python regalis
Habitat: Terrestrial lowland forest
Ecological Role: Prey / Mesopredator
Show more

Recent Activity
May 23, 2025 10:30 AM
Catalog entry created by Dr. Sarah Reyes
Basic info updated by Dr. Sarah Reyes

Recent Cataloging Activity
View all →

Accession No.	Specimen	Updated By	Last Updated	Changes	Status
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
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USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged
USCIB-M-2025-078	Hercules Beetle (Dynastes hercules)	R. de la Cruz	May 23, 2025 10:15 AM	Added habitat, tags	Cataloged

Showing 1 to 5 records of 60 records

Figure 14. Catalogue Page - Location and Media Edit Draft

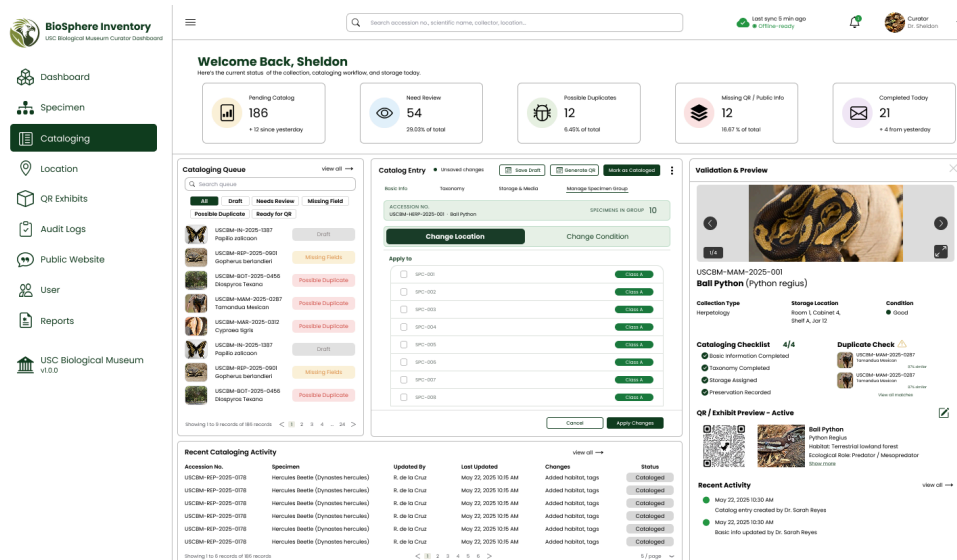


Figure 15. Catalogue Page - Specimen Group Location Edit Draft

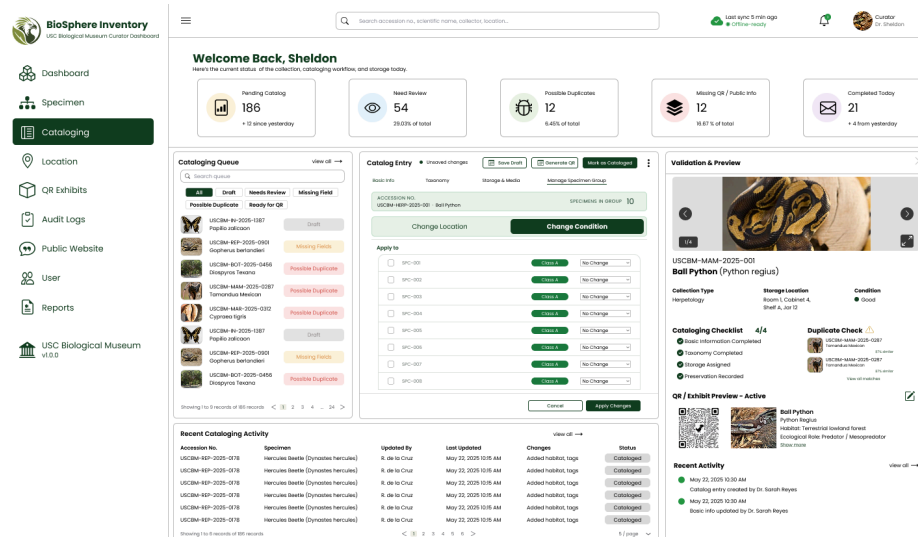


Figure 16. Catalogue Page - Specimen Group Edit Draft

As shown in Figures 17, 18, 19, the Location Management page provides a hierarchical view of storage locations, allowing the curator to navigate from storage rooms to cabinets and drawers while displaying detailed location information, specimen counts, and recent activity. The page uses a three-panel layout with standard functions such as search, add, edit, view, print, and navigation buttons, ensuring an organized, accessible, and efficient user experience.

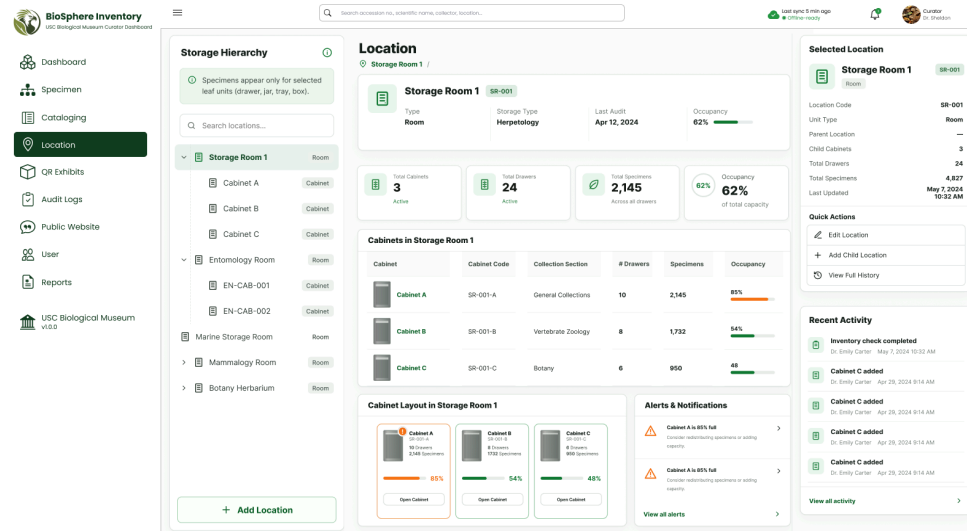


Figure 17. Location Page - Room Draft

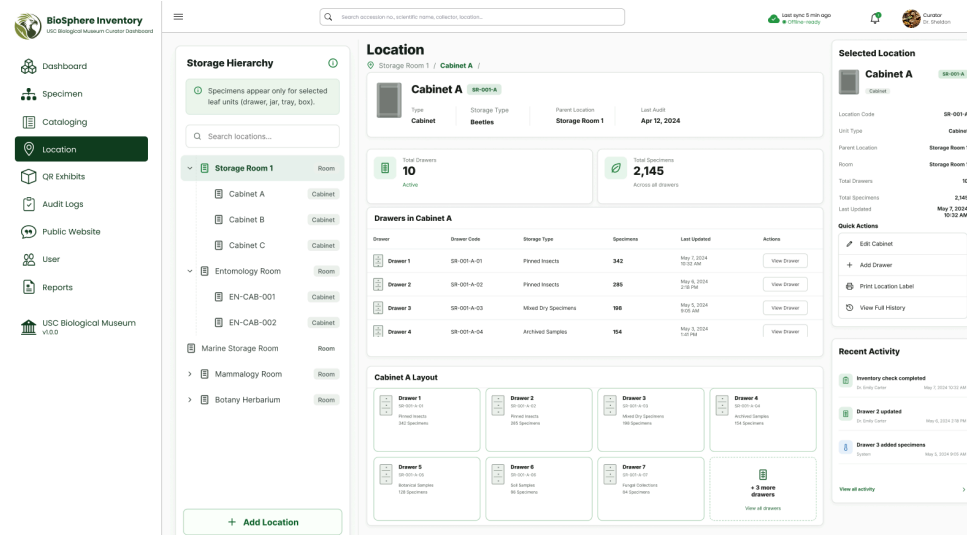


Figure 18. Location Page - Cabinet Draft

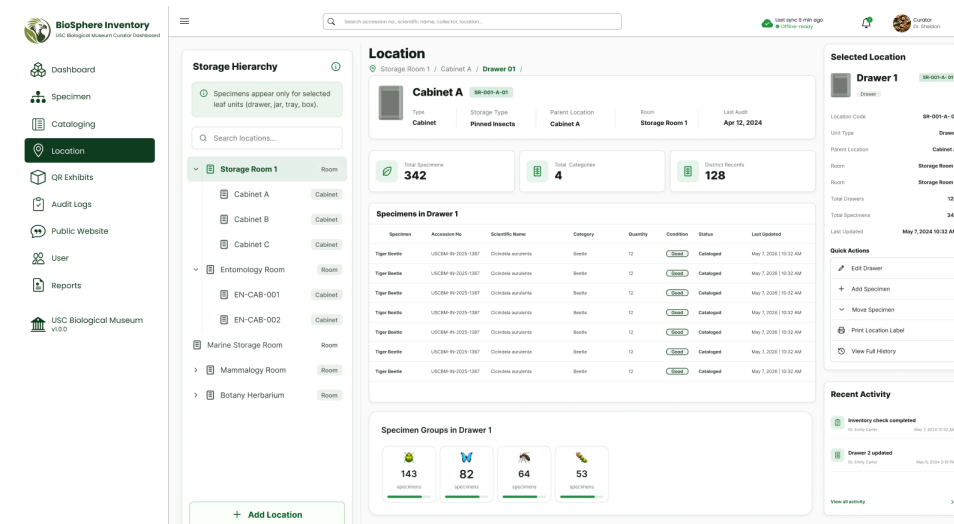


Figure 19. Location Page - Drawer Draft

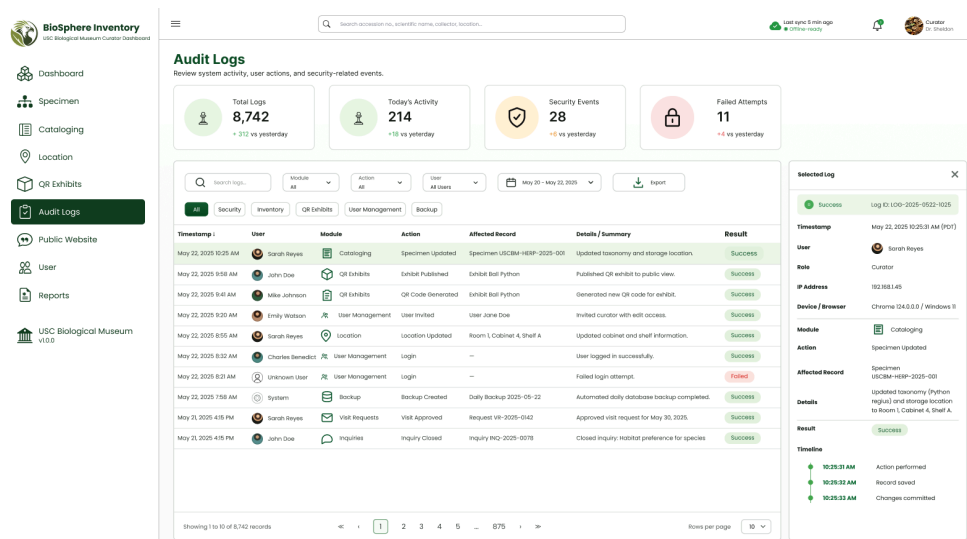


Figure 20. Audit Logs Page Draft

As shown in Figure 20, the Audit Logs page enables curators to review system activities, user actions, and security-related events through a centralized log dashboard. Standard functions include searching and filtering log records, viewing detailed audit information, exporting audit logs, and navigating through log history.

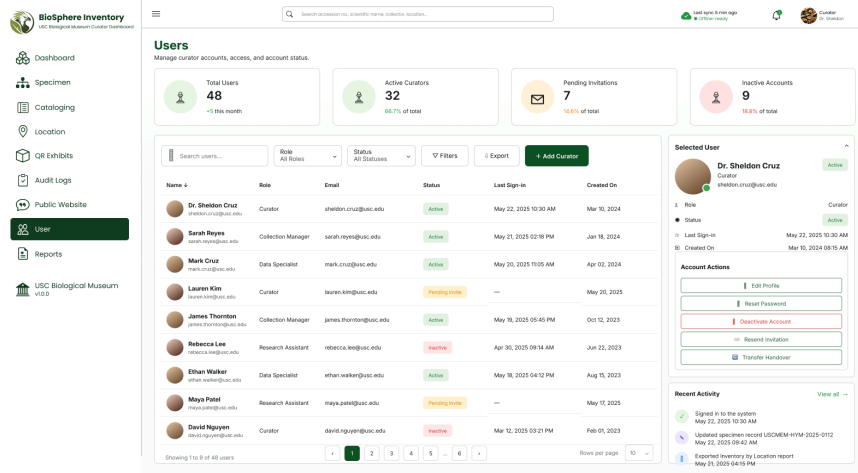


Figure 21. User Management Page Draft

As shown in Figure 21, the User Management page enables the curator to manage curator accounts, monitor account statuses, and view user information from a centralized dashboard. Standard functions include searching and filtering users, adding curator accounts, exporting user records, viewing account details, and performing account management actions such as editing profiles, resetting passwords, and updating account status.

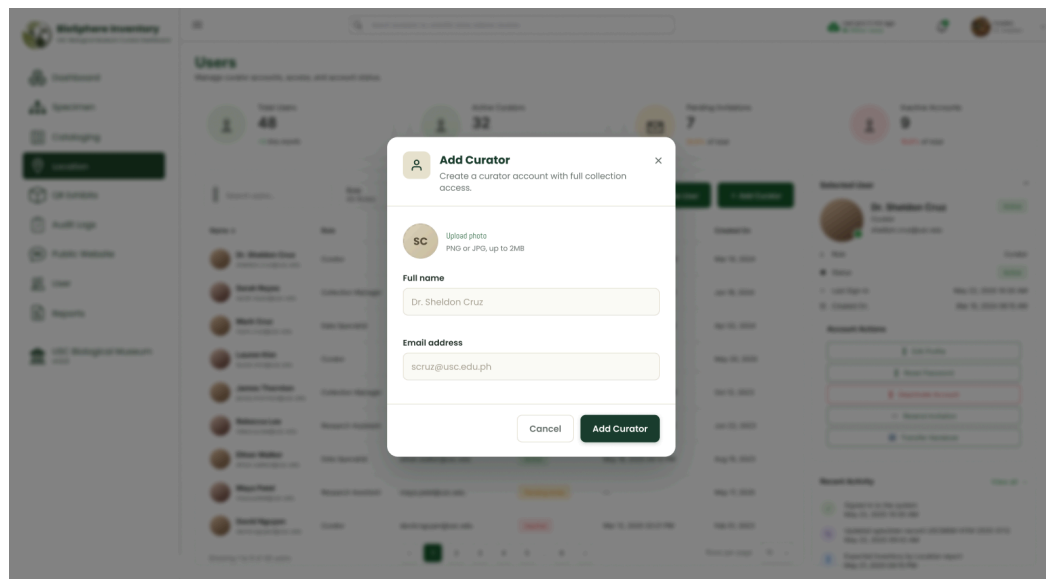


Figure 22. User Management Page - Add User Modal Draft

As shown in Figure 22, the Add Curator modal enables authorized curators to add new curator accounts by entering the required user information.

Standard functions include uploading a profile photo, entering curator details, adding the curator, canceling the operation, or closing the interface.

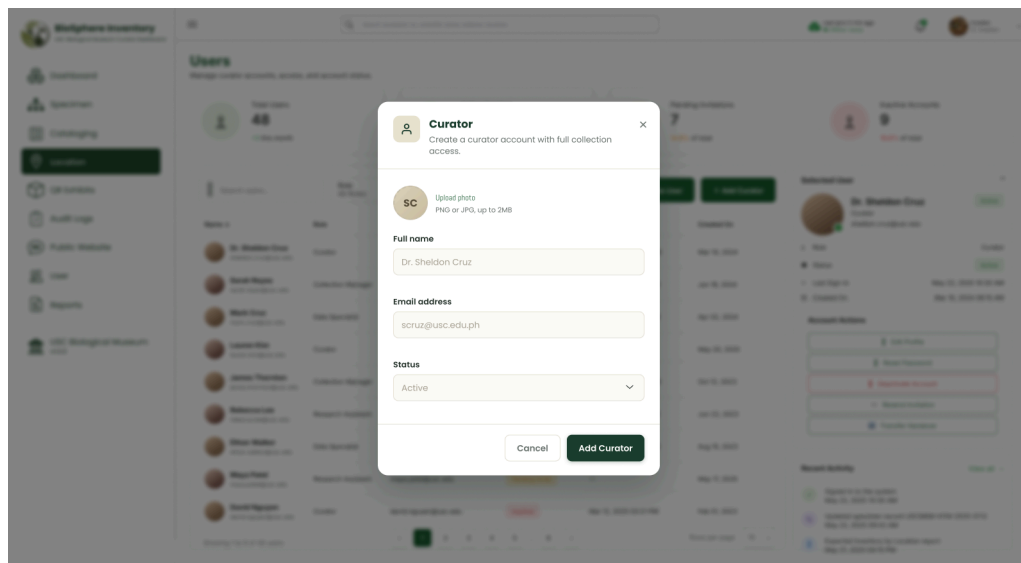


Figure 23. User Management Page - Change User Status Modal Draft

As shown in Figure 23, the Account Status Change modal enables a curator to modify a curator's account status. Standard functions include selecting a new account status, saving the changes, canceling the operation, or closing the interface.

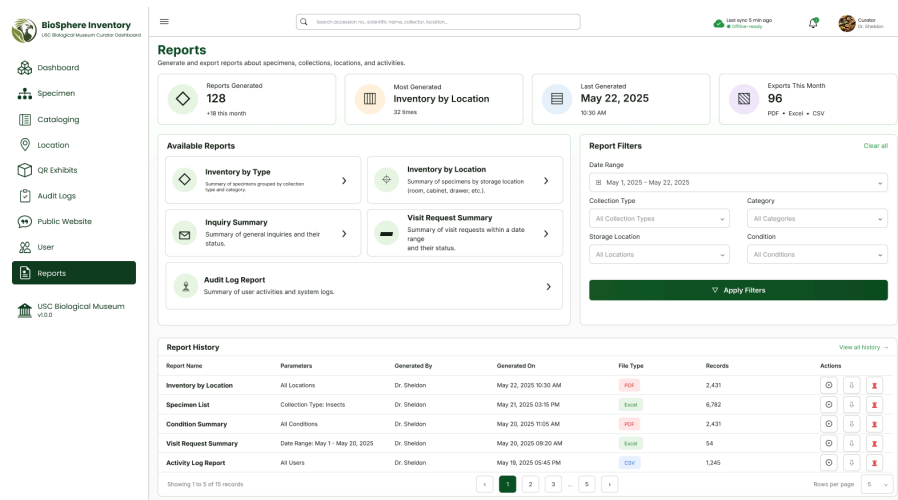


Figure 24. Reports Page Draft

As shown in Figure 24, the Reports page enables curators to generate, filter, view, and export reports related to museum specimens, collections, locations, inquiries, and audit logs. Standard functions include selecting report types, applying filters, viewing report history, exporting reports, and downloading or deleting previously generated reports.

As shown below in Figures 25, 26, 27, the modal enables the curator to export reports by selecting a preferred file format before downloading. Standard functions include choosing a report format (CSV, DOCX, or PDF), confirming the export, canceling the operation, or closing the interface.

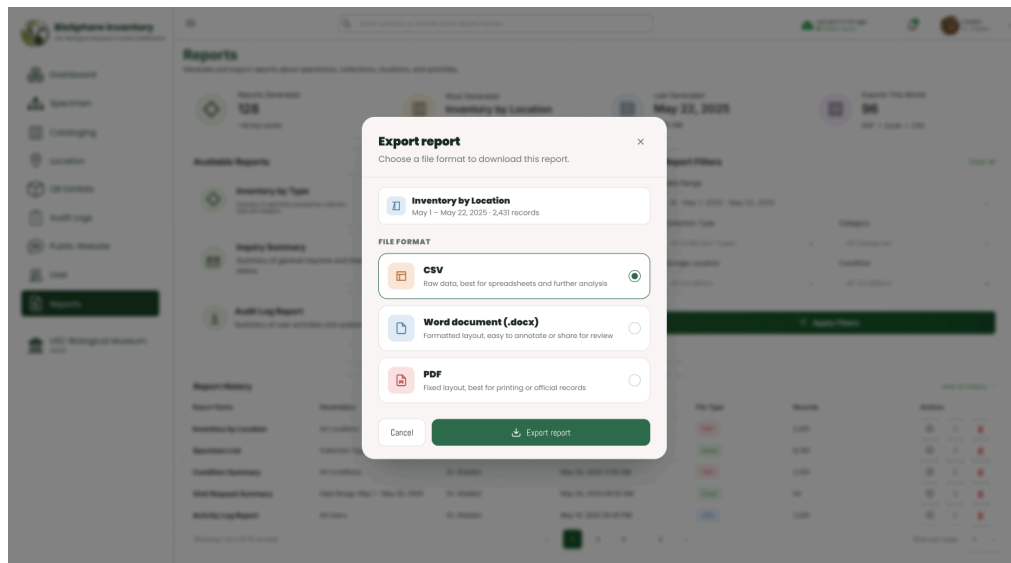


Figure 25. Reports Page - Export as CSV Modal Draft

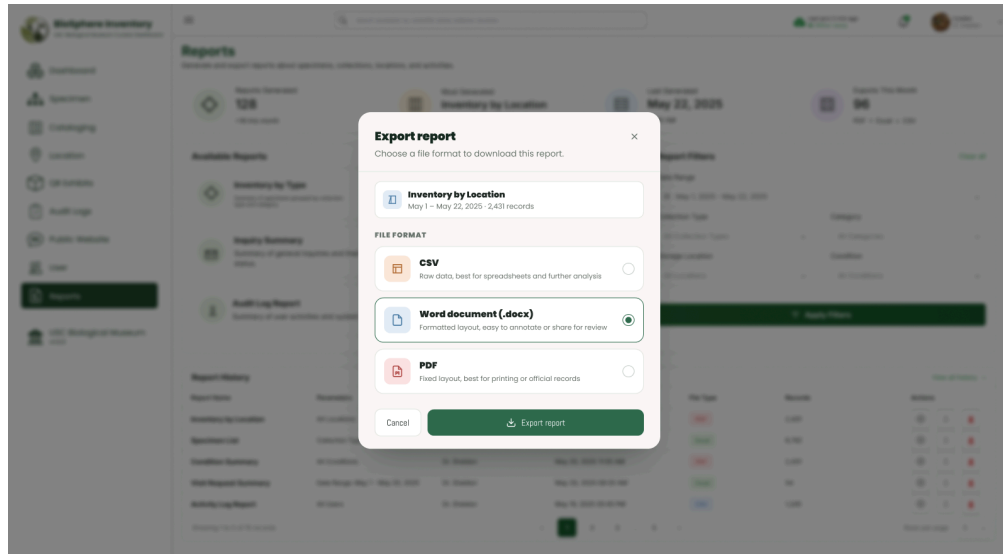


Figure 26. Reports Page - Export as Word Document Modal Draft

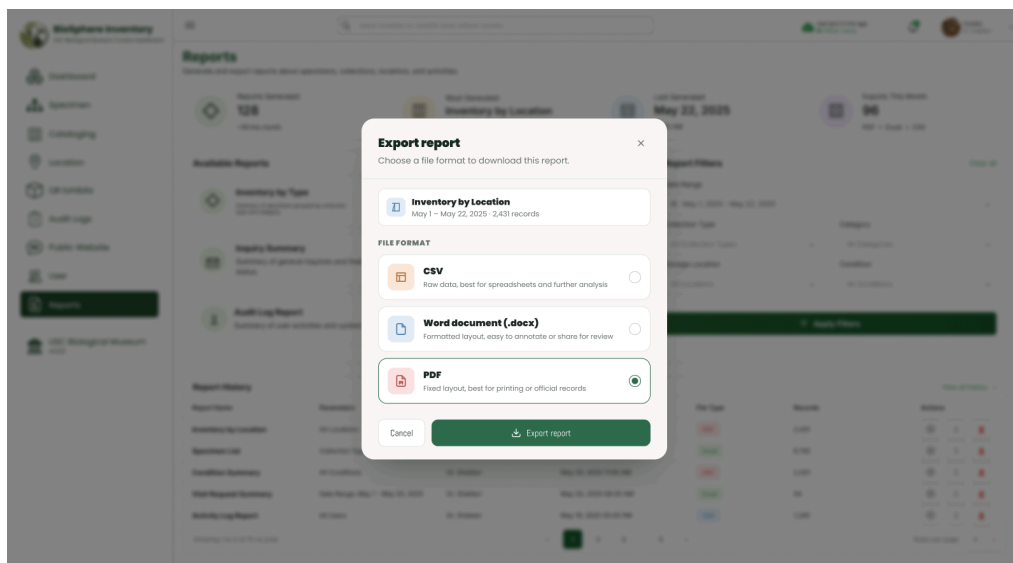


Figure 27. Reports Page - Export as PDF Draft

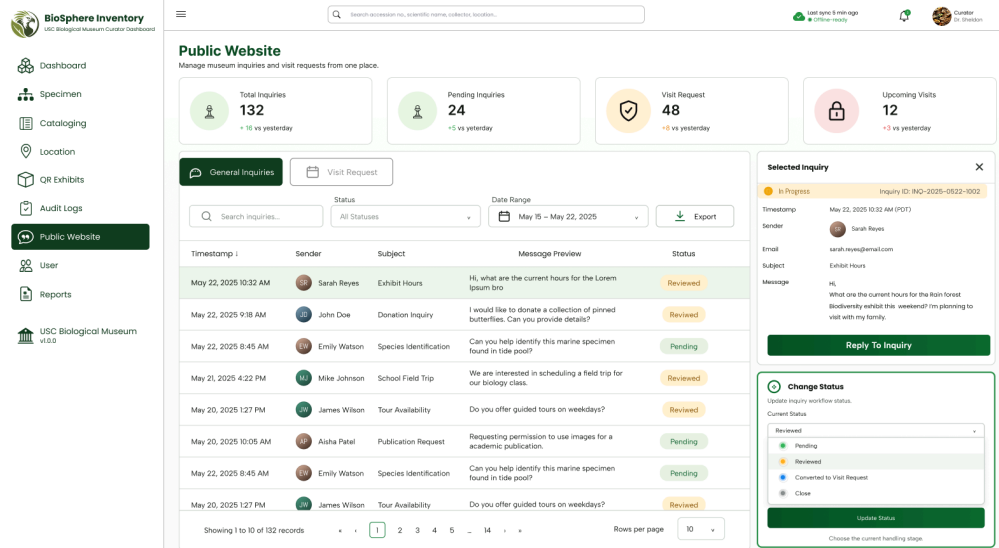


Figure 28. Public Submissions Management Page

As shown in Figure 28, the Public Submissions Management Page displays summary cards (Total Inquiries, Pending Inquiries, Visit Requests, Upcoming Visits) above a tabbed table of submitted General Inquiries and Visit Request forms, filterable by search, status, and date range, with a Reply and Export function. Selecting a record opens a Selected Inquiry panel on the right showing sender details and message content, along with a Reply to Inquiry button and a Change Status control for updating the inquiry's workflow stage (Pending, Reviewed, Converted to Visit Request, Close).

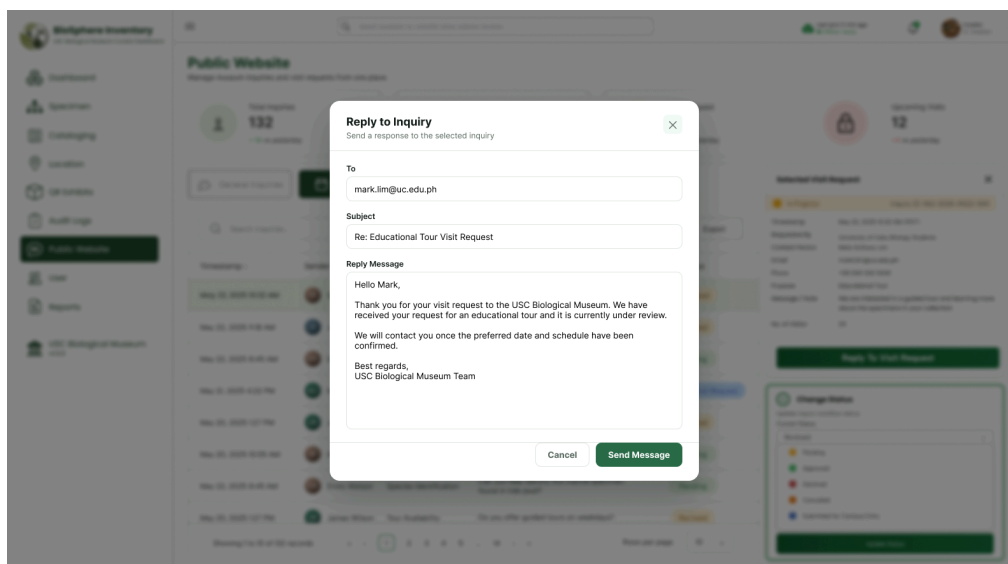


Figure 29. Public Website - Inquiry and Request Email Reply Modal

As shown in Figure 29, the "Reply to Inquiry" modal presents a simple form with To, Subject, and Reply Message fields, along with standard Cancel and Send Message buttons to complete or dismiss the action.

3.1.4 Restricted Developer Interface

The developer interface shall be minimal and shall not reproduce the curator application. A single restricted page or two compact sections may provide: (1) authorized curator account onboarding, activation, disabling, or removal; and (2) approved AR asset upload, replacement, activation, or deactivation. All actions shall require authentication and shall be audited. Routine specimen, location, request, report, FAQ, and public-content functions shall not be visible or accessible.

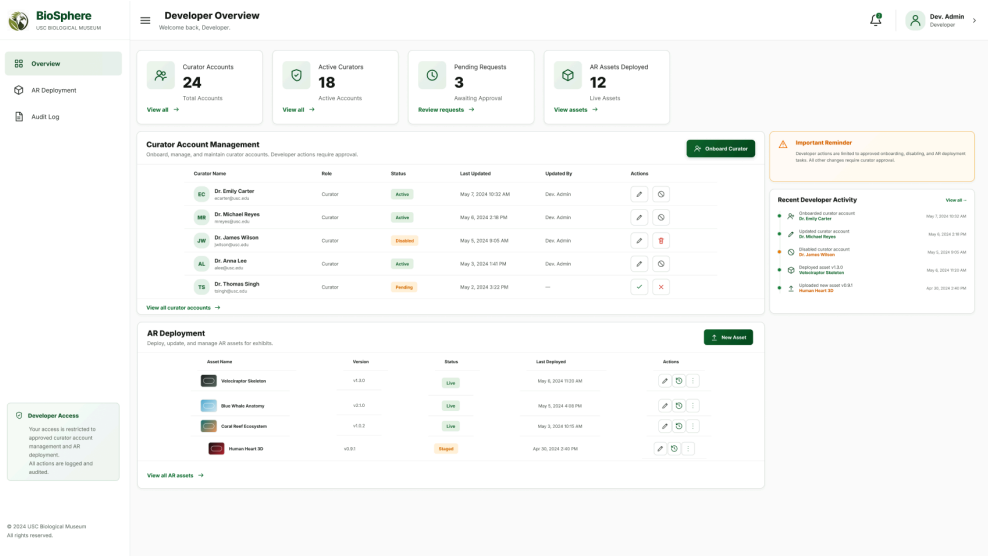


Figure 30. Developer Page draft

As shown in Figure 30, the restricted Developer page provides only authorized curator-access controls and approved AR-asset deployment functions. It does not display visitor requests, specimen records, or routine curator-management modules.

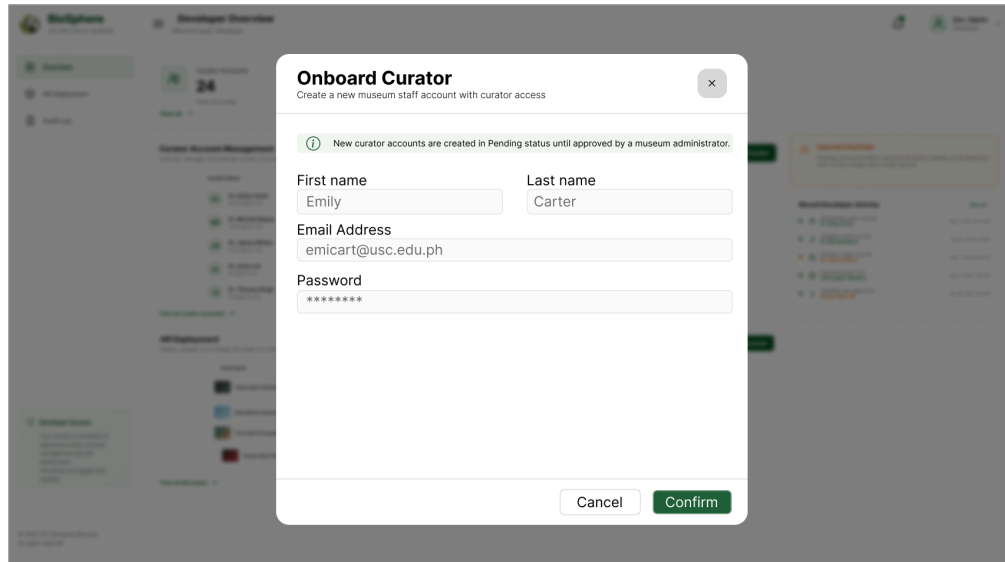


Figure 31. Onboard User Modal draft

As shown in Figure 31, the modal enables developers to create new curator accounts by entering the required user information. Standard functions include providing account details, confirming account creation, canceling the operation, or closing the modal.

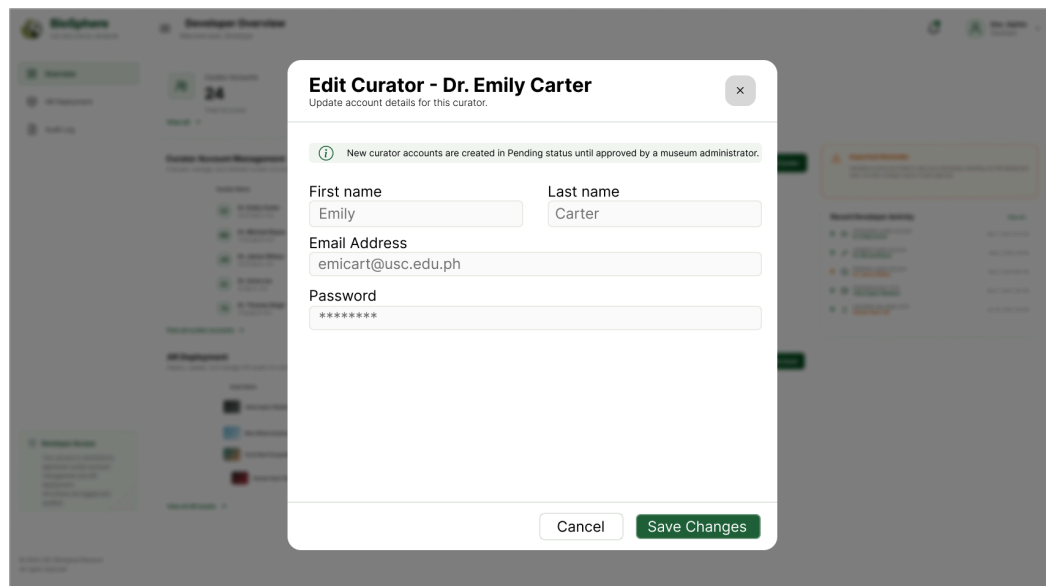


Figure 32. Edit User Modal Draft

As shown in Figure 32, the modal allows the Developer to update authorized curator account details and account status when formally authorized.

The Developer cannot view or assign passwords because password setup and recovery are completed through secure email links.

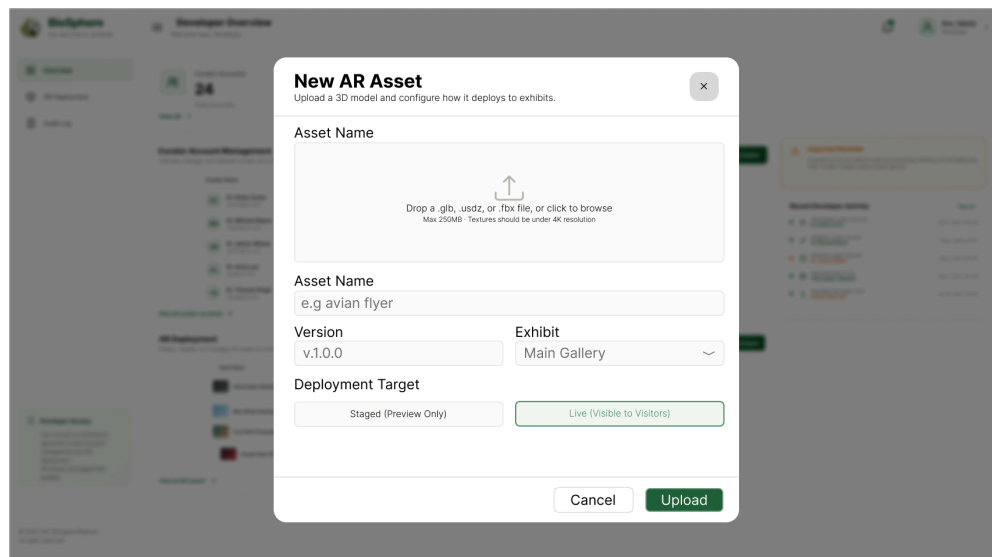


Figure 33. New AR Asset Modal Draft

As shown below in Figure 33, the modal enables developers to upload new AR model files and conFigure their deployment settings. Standard functions include uploading a model file, entering asset details, selecting an exhibit and deployment target, and uploading or canceling the operation.

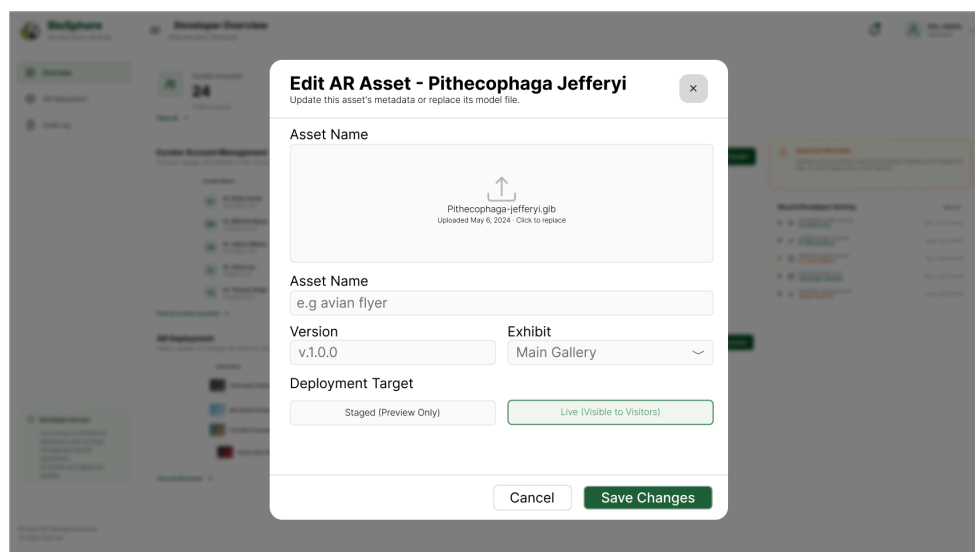


Figure 34. Edit AR Asset Modal Draft

As shown in Figure 34, the modal enables developers to update AR asset information, replace model files, and configure deployment settings. Standard functions include uploading or replacing the model file, editing asset details, selecting an exhibit and deployment target, saving changes, or canceling the operation.

3.1.5 Public QR Exhibit Interfaces

A QR exhibit page shall be optimized for mobile viewing and shall display approved images and educational information through clear expandable sections. The normal exhibit page shall remain available whether or not AR is enabled. Internal fields, condition notes, storage locations, audit information, and curator-only remarks shall never appear.

As shown in Figure 35, the page displays specimen images, taxonomic information, and educational content in a mobile-friendly layout, with standard functions for browsing specimen images, expanding information panels, and scanning another specimen using the Scan Other Specimen button. Figure 36 shows this same page with an added function, a button that launches the 3D AR model view for interactive viewing of the specimen.

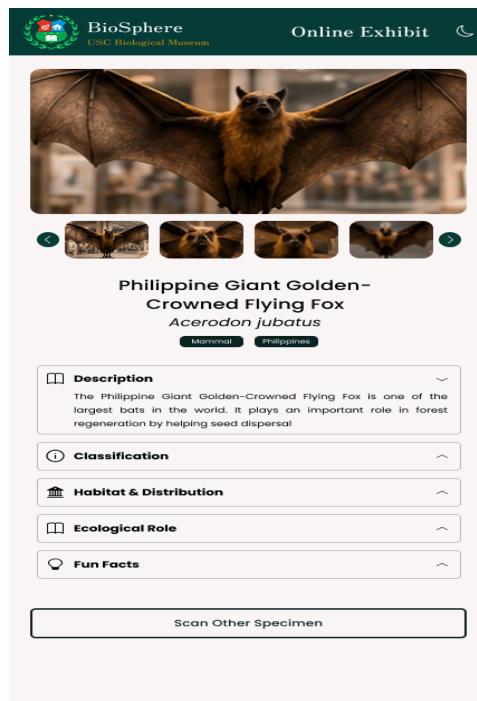


Figure 35. QR exhibit page without an AR action.

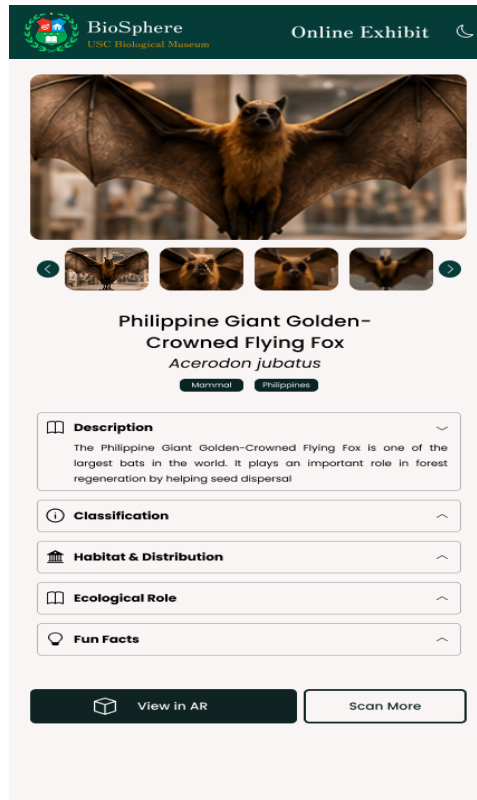


Figure 36. QR exhibit page with the View in AR action

The dark-mode control and an optional Scan Another Specimen convenience action shown in the draft are page enhancements. Camera permission shall be requested only after a visitor explicitly starts an AR or scanning action.

3.1.6 Controlled Knowledge-Based FAQ Assistant Interface

The FAQ assistant shall appear as a public website widget or panel with a clear museum-assistant identity, suggested questions, a message field, a send control, and a visible path to the General Inquiry form. It shall disclose that answers are limited to curator-approved museum knowledge and shall not appear to approve visits or process campus entry.

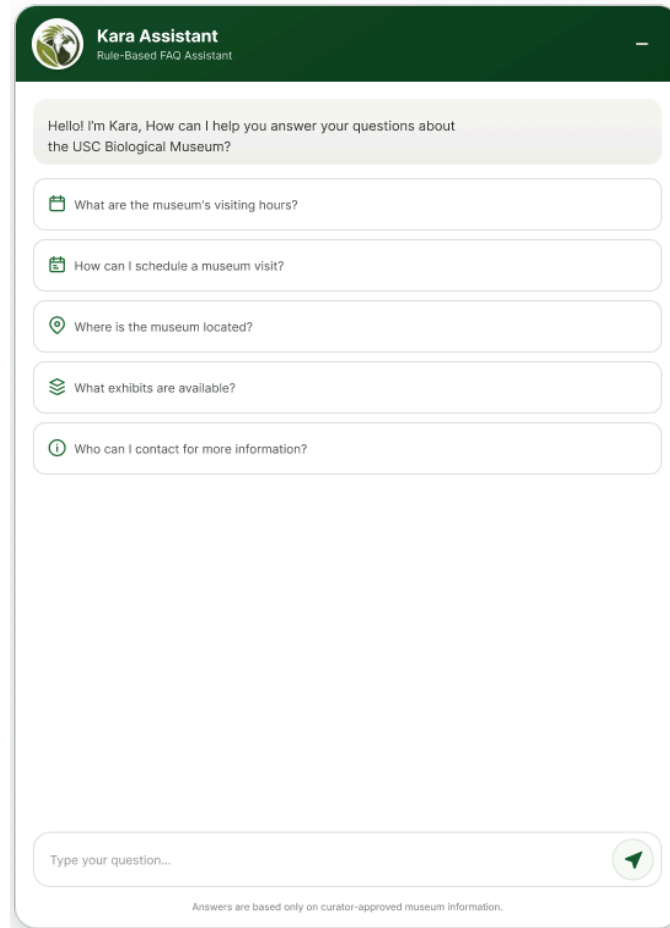


Figure 37. *Controlled Knowledge-Based FAQ Assistant Interface Draft*

As shown in Figure 37, the interface displays a greeting, suggested frequently asked questions, and a text box for custom inquiries. Standard functions include selecting FAQ buttons, submitting questions, minimizing the chatbot, and viewing curator-approved responses.

3.2 Hardware Interfaces

Interface	Purpose	Requirement
Desktop or Laptop Computer	Accesses the curator Progressive Web Application and restricted developer functions.	Keyboard or other text-input device, pointing device, current supported browser, network adapter, and sufficient browser storage for the selected offline data and cached application resources.
Camera-Capable Mobile Device	Opens QR exhibit pages and displays selected browser-based WebAR experiences.	Current supported mobile browser and rear-facing camera. Camera permission is required only when accessing WebAR features that use the camera. Device motion or orientation support is required only when used by the selected WebAR implementation.
Printer (Optional)	Prints QR exhibit labels and generated reports.	Standard operating-system or browser printing support. No dedicated printer model is required, although print

		quality and label size must be suitable for readable QR codes.
Deployment Server	Hosts or supports the public website, curator PWA, backend API, scheduled processes, database access, and related application services.	Final processor, memory, storage, Linux distribution, network configuration, firewall rules, TLS configuration, email-service access, database arrangement, and backup permissions are subject to USC/DCISM infrastructure assessment and approval.
External or Off-Site Backup Storage	Stores encrypted backup copies when approved as part of the backup and recovery procedure.	Approved storage destination, authorized credentials, encryption controls, retention policy, network access, and verified restoration access.

3.3 Software Interfaces

Interface	Purpose	Data Exchanged / Boundary
Next.js public frontend	Provides the public website, Gallery, General Inquiry and	Exchanges REST/JSON data with the backend, retrieves approved public media URLs,

	Visit Request forms, FAQ Assistant, QR exhibit pages, and WebAR launch pages.	and uses supported browser APIs.
Next.js curator PWA	Provides authenticated specimen inventory and museum-management functions.	Exchanges REST/JSON data with the backend and manages cached specimen records, offline-created Uncataloged drafts, basic synchronization status, application resources, and secure session information. It shall not queue offline updates to existing server records.
NestJS backend	Provides centralized business logic, validation, authorization, synchronization, reporting, and audit processing.	Exposes documented REST endpoints using the OpenAPI specification; validates requests, enforces access controls, processes synchronization, generates report outputs, and records audit events.
PostgreSQL + Prisma	Stores and manages persistent relational application data.	Stores curator and developer accounts, specimen records and specimen groups, taxonomy information, storage locations, visit requests, inquiries, FAQ knowledge, exhibits, audit records,

		notifications, and system configuration.
Object storage	Stores approved media and large system assets that are unsuitable for direct database storage.	Stores specimen images and approved three-dimensional or WebAR assets together with file metadata and access-control policies. Generated reports and QR images may be produced on demand unless permanent storage is later approved.
Authentication Service and Backend Authorization Guards	Manages user authentication, secure sessions, and role-based access control.	Exchanges secure session cookies or tokens, user identity, role information, session-expiration data, and authorization results. All protected backend requests remain subject to server-side role and permission checks.
IndexedDB/Dexie + Workbox	Supports selected offline curator operations and application caching.	Stores previously synchronized specimen records for read-only offline access, new Uncataloged text-only drafts awaiting synchronization, basic synchronization metadata, and cached application resources.
Email service	Sends email alerts to the authorized curator for new General Inquiries and Visit	Exchanges the authorized recipient address, Reply-To address, approved subject and message template, relevant

	Requests; sends curator-initiated inquiry replies and Visit Request updates to visitors; and sends approved account-security emails.	inquiry or Visit Request information, delivery status, and secure onboarding or recovery links. The integration is outbound only. BioSphere shall not access external inboxes, retrieve incoming email, detect Gmail replies, import email threads, or synchronize external mailbox content.
QR-Code Generation Library	Generates downloadable or printable QR codes for published exhibit pages.	Receives the published exhibit URL and approved output or label settings and produces a QR image or printable output.
WebAR/model-viewer runtime	Displays approved three-dimensional assets and launches supported browser-based AR experiences.	Uses approved asset URLs, model metadata, AR configuration, browser and device capability information, and camera access when required.
USC ISMIS/campus-entry process	Remains an external manual process used after curator approval of a visit request.	BioSphere displays or exports the necessary approved visit information for authorized personnel. No direct API integration, automatic submission, or modification of ISMIS records is included in the MVP.

3.4 Communications Interfaces

- Production browser-to-server communication shall use HTTPS with a valid TLS configuration.
- The frontends shall communicate with the backend through documented REST/JSON endpoints or an approved equivalent secure API.
- Authentication credentials, session data, visitor personal information, internal inventory changes, and media-upload authorization shall not be transmitted through unencrypted HTTP.
- Email shall be sent through an approved secure email service only for General Inquiry and Visit Request workflows and authorized curator account onboarding, password recovery, and password-change confirmation. Delivery failures shall be logged and surfaced to an authorized user through the appropriate in-system interface.
- BioSphere email integration shall be outbound only. The system shall not connect to, monitor, read, or synchronize an external Gmail, Outlook, or other mailbox. Replies sent by visitors to the configured Reply-To address shall remain outside BioSphere unless the curator manually records the relevant information in the corresponding General Inquiry or Visit Request.
- Synchronization failures, incomplete records, backup events, report generation, specimen changes, location movements, QR or AR activity, and other operational events shall use in-system alerts or audit records and shall not require email delivery.
- Offline synchronization shall submit only new locally created Uncataloged drafts after connectivity is restored and the curator is authenticated. Existing server records shall not be modified through the offline synchronization process.
- Public media access shall follow publication status and access-control rules; private specimen media shall not be exposed through unrestricted public URLs.
- Backup transfers to external storage shall use an approved encrypted transport and access method.

4. System Features

This section defines the major product features in logical operational order. Unless explicitly stated otherwise, High-priority requirements form the MVP baseline.

4.1 Authentication and Curator Account Management

4.1.1 Description and Priority

Description. Authenticates authorized curators and restricted developers, protects user sessions, provides separate credentials, and supports secure email-based curator onboarding, password change, account recovery, activation, deactivation, and succession.

Priority. High

4.1.2 Stimulus/Response Sequences

- An authorized user submits valid credentials for an active account; the system authenticates the user and opens the interface permitted for the assigned role.
- A user submits invalid credentials, attempts to use an inactive account, or fails an applicable authentication check; the system denies access and records the event according to the approved security-logging policy.
- An authorized curator initiates the creation of an account for another authorized curator; the system creates a separate account using the approved onboarding procedure and records the account-management action.
- An authorized curator initiates an account handover; the successor receives a separate authorized account, while the outgoing curator's account is retained, deactivated, or updated according to the approved handover procedure without sharing credentials.
- A curator logs out or the session expires; the system terminates access and requires reauthentication.

- An authorized curator or Developer initiates approved curator onboarding; the system creates the pending authorized account and sends the new curator a secure email invitation to establish a password.
- A curator selects Change Password while authenticated; the system verifies the applicable authentication requirements, changes the password, terminates or updates affected sessions according to policy, records the event, and may send a password-change confirmation email.
- A curator requests password recovery; the system displays a generic confirmation response and sends a secure, time-limited password-reset link when the submitted address belongs to an eligible account.
- An onboarding or recovery link is invalid, expired, or already used; the system rejects it without exposing account or credential information.

4.1.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.1-01	The system shall authenticate registered and active curator accounts before granting access to curator functions.	High
REQ-4.1-02	The system shall provide each curator with separate credentials and shall not require credential sharing.	High
REQ-4.1-03	The system shall prevent public self-registration for curator accounts.	High

REQ-4.1-04	The system shall create a protected authenticated session after successful login.	High
REQ-4.1-05	The system shall provide secure logout and terminate the session after logout or approved inactivity timeout.	High
REQ-4.1-06	The system shall prevent unauthenticated users from accessing protected curator or developer routes, restricted data, and protected API operations.	High
REQ-4.1-07	The system shall allow an authorized curator to initiate the creation of a separate account for another authorized curator through the approved onboarding procedure and shall send the new curator a secure password-setup invitation email.	High
REQ-4.1-08	The system shall allow an authorized curator to activate or deactivate another curator account only according to the approved museum account-management and handover procedure.	High
REQ-4.1-09	The system shall support curator succession or handover by providing the successor with a separate authorized account without transferring or sharing the outgoing curator's credentials.	High
REQ-4.1-10	The system shall provide an approved email-based account-recovery or password-reset process using a secure, time-limited link and shall not expose the account's current password, temporary password, recovery token, or account-existence status.	High

REQ-4.1-11	The system shall record curator account creation, onboarding invitation, password setup, password change, activation, deactivation, account recovery, password reset, and handover actions in the audit log without recording passwords, invitation secrets, reset tokens, or session secrets.	High
REQ-4.1-12	The system shall prevent inactive curator accounts from authenticating or receiving new operational permissions.	High
REQ-4.1-13	The system shall authenticate registered and active Developer accounts before granting access to restricted Developer functions.	High
REQ-4.1-14	The system shall enforce role-based access controls that prevent curators from accessing Developer-restricted functions and prevent Developers from accessing routine curator operations.	High
REQ-4.1-15	The system shall record successful and unsuccessful authentication attempts and restricted account-management actions according to the approved security-logging policy without recording passwords, session secrets, or recovery tokens.	High
REQ-4.1-16	The system shall allow an authenticated curator or Developer to change their own password after completing the approved authentication checks.	High
REQ-4.1-17	The system shall not send a current or temporary password through email during onboarding, password change, or account recovery.	High

REQ-4.1-18	The system should send a confirmation email to the account holder after a successful password change, without including the password or authentication secrets.	Medium
REQ-4.1-19	The system shall ensure that onboarding invitations and password-reset links expire after the approved period and cannot be reused after successful completion.	High

4.2 Restricted Developer Functions

4.2.1 Description and Priority

Description. Provides only the restricted technical capabilities needed for initial curator account provisioning, formally authorized curator-account activation or deactivation, and deployment or maintenance of approved AR assets.

Priority. High

4.2.2 Stimulus/Response Sequences

- An authorized developer signs in; the system opens only the restricted technical interface.
- The Developer provisions the initial curator account or performs a formally authorized curator-account activation or deactivation; the system applies the approved change and records the action in the protected audit log.
- The Developer uploads, replaces, or activates an approved AR asset for an exhibit previously approved by the Museum Curator; the system validates the asset, associates it with the approved exhibit, and records the deployment action.
- The Developer attempts to access specimen, location, inquiry, visit-request, report, FAQ-knowledge, Gallery-content, or other routine curator-management functions; the system denies access and records the unauthorized attempt according to the security-logging policy.

4.2.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.2-01	The system shall authenticate the developer before granting access to restricted technical functions.	High
REQ-4.2-02	The system shall allow an authorized Developer to provision the initial curator account through the approved onboarding procedure and trigger a secure password-setup invitation email to the authorized curator.	High
REQ-4.2-03	The system shall allow an authorized Developer to activate or deactivate a curator account only when formal authorization is recorded.	High
REQ-4.2-04	The system shall allow a developer to upload, replace, activate, deactivate, or remove approved AR assets linked to selected exhibits.	High
REQ-4.2-05	The system shall validate the AR asset's file type, file size, required metadata, technical compatibility, and documented authorization for use before activation.	High
REQ-4.2-06	The system shall prevent the developer from creating, editing, archiving, moving, or publishing specimen records.	High
REQ-4.2-07	The system shall prevent the developer from managing routine locations, inquiries, visit requests, reports, FAQ knowledge, and public website content.	High

REQ-4.2-08	The Developer interface shall provide only authorized curator-account controls and approved AR-asset deployment functions and shall not expose routine curator-management modules.	High
REQ-4.2-09	The system shall audit all developer account and AR deployment actions.	High

4.3 Curator Dashboard and Notifications

4.3.1 Description and Priority

Description. Provides a concise operational overview, quick access to common tasks, in-system alerts, and connection/synchronization status.

Priority. High

4.3.2 Stimulus/Response Sequences

- The curator opens the PWA; the system displays inventory summaries, pending visit requests, unresolved inquiries, recent in-system alerts, connection status, and synchronization state.
- The curator selects a summary card; the system opens the corresponding filtered module.
- A new General Inquiry or Visit Request is submitted; the system creates an in-system alert and sends an email notification to the authorized curator.
- Synchronization of an offline-created draft fails; the dashboard retains the draft, displays a Sync Failed state, and allows the curator to review the failure information and retry after connectivity is available.

4.3.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.3-01	The system shall display counts for Uncataloged, Cataloged, and Archived specimen records and selected collection categories.	High
REQ-4.3-02	The system shall display pending general inquiries and visit requests requiring curator attention.	High
REQ-4.3-03	The system shall display Online, Offline, Pending Sync, Synchronizing, Synchronized, and Sync Failed states for offline-created specimen drafts.	High
REQ-4.3-04	The system shall display quick actions for adding a specimen, reviewing requests, opening reports, and reviewing offline-created drafts awaiting synchronization.	High
REQ-4.3-05	The system shall provide in-system notifications for new inquiries, new visit requests, sync failures, and other curator-approved operational alerts.	High
REQ-4.3-06	The system shall allow the curator to open the related record or module from a notification.	High
REQ-4.3-07	The system shall allow the curator to mark notifications as read or dismissed without deleting or modifying the underlying audit or operational record.	Medium
REQ-4.3-08	The system shall not expose private visitor or specimen details in a notification shown to an unauthorized or unauthenticated user.	High

REQ-4.3-09	The system shall send operational email alerts to the authorized curator when a new General Inquiry or Visit Request is submitted.	High
REQ-4.3-10	The system shall use in-system notifications rather than email for synchronization failures, incomplete records, backup events, report results, specimen changes, storage-location changes, QR activity, AR activity, and other internal operational alerts.	High

4.4 Specimen Inventory and Cataloging

4.4.1 Description and Priority

Description. Creates and maintains the museum's master specimen inventory, including classification and descriptive information, associated images, record states, import validation, and possible duplicate-record warnings.

Priority. High

4.4.2 Stimulus/Response Sequences

- The curator manually creates a record; the system assigns a unique system-generated specimen ID and evaluates completeness.
- A record is saved with missing applicable required non-image fields; the system saves it as Uncataloged and identifies the missing fields.
- The curator completes all applicable required non-image fields or approved Unknown/Not Applicable values; the system classifies the record as Cataloged.
- The curator imports a spreadsheet or CSV file; the system validates each row, identifies errors and possible duplicate records, and allows valid records to be reviewed before they are committed to the inventory.
- The curator archives an inactive record; the system removes it from active inventory and public eligibility while preserving its history.

- The curator edits an existing specimen record; the system validates the changes, updates the applicable record state, and records the modification in the audit log.

4.4.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.4-01	The system shall allow the curator to create and save a specimen record.	High
REQ-4.4-02	The system shall automatically assign every specimen record a unique system-generated specimen ID.	High
REQ-4.4-03	The system shall allow the museum accession number or code to remain null until assigned or confirmed.	High
REQ-4.4-04	The system shall enforce the curator-approved accession-number uniqueness rule after the official format, scope, and assignment procedure have been validated.	High
REQ-4.4-05	The system shall assign each specimen record the status Uncataloged, Cataloged, or Archived.	High
REQ-4.4-06	The system shall classify a newly saved record as Uncataloged when at least one applicable required non-image field is incomplete.	High
REQ-4.4-07	The system shall classify a record as Cataloged only when all applicable required non-image fields contain valid information or an approved Unknown or Not Applicable value.	High

REQ-4.4-08	The system shall identify and display missing or invalid applicable required fields.	High
REQ-4.4-09	The system shall allow the curator to view and edit Uncataloged and Cataloged records.	High
REQ-4.4-10	The system shall archive inactive specimen records instead of permanently deleting them through routine curator operations.	High
REQ-4.4-11	The system shall prevent Uncataloged and Archived records from being published as public exhibit content.	High
REQ-4.4-12	The system shall allow the curator to attach, view, replace, and remove supported specimen images.	High
REQ-4.4-13	Specimen images shall be optional and shall not prevent an otherwise complete record from becoming Cataloged.	High
REQ-4.4-14	The system shall store classification status and condition information only as entered or approved by an authorized curator.	High
REQ-4.4-15	The system shall allow the curator to mark or unmark a Cataloged specimen record as eligible for public exhibit use through a public-display flag.	High
REQ-4.4-16	The public-display flag shall not automatically publish the complete internal specimen record.	High
REQ-4.4-17	The system shall preserve the record creator, last editor, creation date and time, last update date and time, and, when applicable, the curator who archived the record and the archive date and time.	High

REQ-4.4-18	The system shall allow manual specimen record encoding.	High
REQ-4.4-19	The system shall allow import through an approved spreadsheet or CSV template.	High
REQ-4.4-20	The system shall validate imported rows and identify missing, invalid, or unmapped values before final saving.	High
REQ-4.4-21	The system shall warn the curator when a manually entered or imported record may duplicate an existing record.	High
REQ-4.4-22	The system shall not automatically merge or delete possible duplicates; the curator shall decide how they are handled.	High
REQ-4.4-23	The system shall allow separate records for the same species when they represent different physical groups, collectors, donors, storage assignments, or other curator-approved distinctions.	High

4.4.4 Supported Specimen Data

Data Group	Supported Information
Identification	System-generated specimen ID; nullable accession number or code; specimen category; scientific name; common name; and provisional identification where curator-approved.
Taxonomy	Kingdom, phylum, class, order, family, genus, and species where available.

Acquisition and collection	Collector, donor, collection date, collection location, acquisition-related notes, remarks, and other available curator-validated provenance information.
Preservation and storage	Preservation method; one or more specimen lots; storage unit per lot; condition class per lot; quantity per lot; calculated total quantity; storage notes.
Public educational information	Habitat, ecological role, conservation status, educational description, interesting facts, approved images, and public-display approval.
System fields	Specimen status, classification status, public-display flag, created by, updated by, creation and update timestamps, archive information, import source, and specimen-lot history.

4.5 Specimen Lot Grouping and Quantity Management

4.5.1 Description and Priority

Description. Represents grouped physical specimens through one or more specimen lots so that the quantity, condition, or location of part of a group can change while preserving an accurate calculated total quantity for the specimen record.

Priority. High

4.5.2 Stimulus/Response Sequences

- A record has quantity 10 in one location and condition A; the system represents it as a specimen lot with quantity 10.
- The curator moves one item to another drawer; the system reduces the source lot to 9 and creates or updates a target lot with quantity 1.
- The curator changes the condition of one item from Condition A to Condition B; the system reduces the Condition A source lot from 10 to 9

and creates or updates a Condition B lot with quantity 1, while the specimen record's calculated total quantity remains 10.

- A target lot already exists with the same specimen, location, and condition; the system adds the selected quantity to the existing lot.
- The curator records an authorized quantity adjustment, such as an addition, loss, transfer, removal, destruction, or correction; the system updates the affected lot and calculates total quantity only after recording the adjustment type, quantity, reason, and responsible curator.
- The curator enters a quantity greater than the available source-lot quantity or a value less than one; the system rejects the operation and displays a validation message.

4.5.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.5-01	The system shall allow one specimen record to contain one or more active specimen lots.	High
REQ-4.5-02	Each specimen lot shall have exactly one storage location, one condition class, and a positive quantity.	High
REQ-4.5-03	The system shall calculate the specimen record total quantity as the sum of all active specimen-lot quantities.	High
REQ-4.5-04	When only part of a lot is moved, the system shall reduce the source lot by the selected quantity and create or update a target lot at the new location.	High
REQ-4.5-05	When only part of a lot changes condition, the system shall reduce the source lot and create or update a target lot with the new condition.	High

REQ-4.5-06	A partial location or condition change shall not change the total quantity of the parent specimen record.	High
REQ-4.5-07	When a target lot already exists with the same specimen, location, and condition, the system shall merge the affected quantity into that lot instead of creating a duplicate lot.	High
REQ-4.5-08	The system shall prevent a move or condition change from using a quantity greater than the source lot quantity.	High
REQ-4.5-09	The system shall change the overall specimen quantity only through an authorized addition, removal, transfer out, deaccession, missing/loss, destruction, or data correction.	High
REQ-4.5-10	The system shall require or record a reason for quantity removals, condition changes, and location changes when applicable.	High
REQ-4.5-11	The system shall record the acting curator, timestamp, source lot, target lot, and quantity affected for each lot split or merge.	High

4.6 Hierarchical Storage-Location Management

4.6.1 Description and Priority

Description. Maintains the museum's flexible physical storage hierarchy and supports specimen-lot assignment, container relocation, hierarchical location tracking, and movement history.

Priority. High

4.6.2 Stimulus/Response Sequences

- The curator creates a room, gallery, cabinet, drawer, shelf, box, vial, or other approved storage unit and assigns it to a valid parent unit, except for an authorized top-level room or gallery.
- The curator opens a cabinet or drawer; the system displays its child units and contained specimen lots.
- The curator assigns or moves a specimen lot to a valid storage unit; the system updates the location and records the movement.
- The curator moves a container; the system changes the container parent while preserving its child hierarchy.
- A user attempts to assign a lot to a unit that cannot hold specimens; the system rejects the assignment.
- The curator attempts to assign a storage unit to itself, one of its descendants, or an incompatible parent type; the system rejects the change and preserves the existing hierarchy.

4.6.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.6-01	The system shall support rooms or galleries, cabinets, drawers or shelves, boxes, vials, and other curator-approved storage-unit types.	High
REQ-4.6-02	The system shall allow the curator to create, view, update, archive, and organize storage units.	High
REQ-4.6-03	The system shall use a parent-child relationship so that a storage unit may be assigned to a valid parent storage unit when applicable.	High
REQ-4.6-04	The system shall store a unit label, unit type, parent, holds-specimens setting, and optional capacity for each storage unit.	High

REQ-4.6-05	The system shall allow a specimen lot to be assigned only to a storage unit configured to hold specimens.	High
REQ-4.6-06	The system shall allow the curator to move all or a selected quantity of a specimen lot to another valid storage unit.	High
REQ-4.6-07	The system shall allow the curator to move a container by changing its parent while preserving its child structure.	High
REQ-4.6-08	The system shall derive the higher-level room or gallery from the assigned storage hierarchy instead of duplicating inconsistent location fields.	High
REQ-4.6-09	The system shall record each specimen-lot location movement with the affected quantity, previous location, new location, acting curator, date and time, and movement reason.	High
REQ-4.6-10	The system shall prevent detailed internal storage locations from appearing on the public website, QR pages, FAQ responses, and AR experiences.	High
REQ-4.6-11	The system shall warn the curator when assigning specimens would exceed an optional configured storage-unit capacity.	Medium
REQ-4.6-12	The system shall prevent a storage unit from being assigned to itself or to one of its descendants and shall reject invalid parent-child combinations.	High

4.7 Search, Filtering, Reports, and Export

4.7.1 Description and Priority

Description. Enables reliable retrieval of museum records and produces the approved operational reports and outputs needed for inventory control, visitor coordination, and turnover.

Priority. High

4.7.2 Stimulus/Response Sequences

- Enables reliable searching and filtering of museum records and generates the five approved operational reports and supported outputs needed for inventory control, visitor coordination, auditing, and turnover.
- The curator applies taxonomy, status, condition, location, or publication filters; the system updates the result set.
- The curator selects a report type, period, filters, and output format; the system generates the report and records the action.
- The system cannot generate the requested report; it displays an error, records the failed operation when applicable, and does not present or export an incomplete report as successful.

4.7.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.7-01	The system shall allow search by accession number, system specimen ID, scientific name, common name, category, and other supported identifiers.	High
REQ-4.7-02	The system shall allow filtering by taxonomy, specimen category, specimen status, classification status, condition, public-display status, and storage location.	High

REQ-4.7-03	The system shall allow supported inventory tables to be sorted by relevant columns.	High
REQ-4.7-04	The system shall allow the curator to select a report type, reporting period, applicable filters, and supported output format.	High
REQ-4.7-05	The system shall generate a Consolidated Museum Operations Report for a monthly, yearly, or custom period.	High
REQ-4.7-06	The Consolidated Museum Operations Report shall include inventory summaries, inventory changes, condition/category breakdowns, inquiry summaries, visit-request summaries, QR/AR summaries, and curator remarks where applicable.	High
REQ-4.7-07	The system shall generate a complete or filtered Inventory Report by status, category, taxonomy, condition, location, public-display status, and date added.	High
REQ-4.7-08	The system shall generate a General Inquiry Summary by date range and status.	High
REQ-4.7-09	The system shall generate a Visit-Request Summary by date range and status.	High
REQ-4.7-10	The system shall generate a QR and AR Exhibit Report covering published, unpublished, disabled, and AR-enabled exhibit pages.	High
REQ-4.7-11	The system shall allow the curator to export the Consolidated Museum Operations Report in DOCX format for offline editing.	High

REQ-4.7-12	The system shall provide PDF or printable output for supported reports and CSV output for applicable tabular data exports.	Medium
REQ-4.7-13	The system shall include private visitor information and detailed internal storage-location data only in authorized internal reports and exports and shall prevent such information from appearing in public outputs.	High
REQ-4.7-14	The system shall record report generation, filters, output type, acting curator, and timestamp in the audit log.	High
REQ-4.7-15	The system shall validate report parameters and display an understandable failure message when generation cannot be completed.	High
REQ-4.7-16	The system shall generate editable DOCX reports as separate exported files, and changes made to an exported file shall not automatically modify the source records stored in BioSphere.	High

4.7.4 Approved Core Report Types

#	Report	Scope	Output
1	Consolidated Museum Operations Report	Inventory summaries and changes, condition and category breakdowns, General Inquiries, Visit Requests, QR and AR exhibits, and curator remarks	Editable DOCX required; PDF and printable output supported.

		for the selected reporting period.	
2	Inventory Report	Complete or filtered specimen inventory.	CSV for detailed tabular data; PDF and printable output for formatted reports.
3	General Inquiry Summary	Inquiry counts/details by period and status.	PDF and printable output; CSV for applicable inquiry details and summary data.
4	Visit-Request Summary	Visit requests by period and status.	PDF and printable output; CSV for applicable visit-request details and summary data.
5	QR and AR Exhibit Report	Publication and AR availability summary.	PDF and printable output; CSV for applicable exhibit and

			AR-status data.
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4.8 General Inquiry Management

4.8.1 Description and Priority

Description. Receives public museum inquiries, supports curator review, curator-initiated outbound email replies and status tracking, and allows an authorized curator to manually refer an appropriate inquiry to the formal Visit Request workflow. Incoming replies from external email accounts are outside BioSphere.

Priority. High

4.8.2 Stimulus/Response Sequences

- A visitor submits a valid inquiry without an account; the system stores it, confirms submission, and alerts the curator.
- The curator opens an inquiry, reviews it, composes an outbound reply through BioSphere, and changes its status; the system sends the message to the visitor's email address and records the delivery result.
- The visitor replies through the museum's external email address; BioSphere does not automatically detect or import the reply, and the curator may manually update the inquiry or add an internal note when necessary.
- The curator determines that an inquiry should proceed as a Visit Request; the system allows the curator to initiate the manual referral, marks the inquiry as Turned to Visit Request, and does not automatically approve the resulting request.
- The form contains invalid or missing required data; the system prevents submission and identifies the issue.

4.8.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.8-01	The public website shall allow a visitor to submit a general museum inquiry without creating an account.	High
REQ-4.8-02	The inquiry form shall collect only the approved minimum contact and message information needed for processing.	High
REQ-4.8-03	The inquiry form shall validate required fields, input lengths, and email format before submission.	High
REQ-4.8-04	The system shall store each inquiry with a unique identifier, submission timestamp, status, and consent/privacy record.	High
REQ-4.8-05	The system shall allow the curator to view, search, filter, and compose outbound email replies for General Inquiries through BioSphere.	High
REQ-4.8-06	The system shall support the General Inquiry statuses Pending, Reviewed, Turned to Visit Request, and Closed.	High
REQ-4.8-07	The system shall allow the curator to manually refer an inquiry to the visit-request workflow.	High
REQ-4.8-08	The system shall send the authorized curator an in-system alert and email notification when a new General Inquiry is submitted.	High
REQ-4.8-09	The system shall send curator-initiated outbound replies through BioSphere to the visitor's email	High

	address and record the message, sending time, acting curator, and available delivery result.	
REQ-4.8-10	The system shall not expose inquiry information to public users or other unauthorized actors.	High
REQ-4.8-11	The system shall record curator-initiated outbound replies, status changes, referral actions, and manually entered internal notes with the acting curator and date and time. The system shall not automatically record or import replies received through an external email mailbox.	High

4.9 Visit-Request Management

4.9.1 Description and Priority

Description. Receives structured Visit Requests, records one or more preferred schedules and optional vehicle information, supports curator decisions and visitor updates, and preserves the external USC campus-entry boundary.

Priority. High

4.9.2 Stimulus/Response Sequences

- A visitor submits a Visit Request with valid required information and one or more preferred schedules; the system stores it with a Pending status, confirms submission, and alerts the curator.
- The curator reviews the request and may compose an outbound request for additional information through BioSphere, which sends the message to the visitor's email address and records the delivery result.
- The visitor may respond through the museum's official email address outside BioSphere. The system does not automatically detect or import the response; the curator may manually update the Visit Request or record an internal note based on the received information.

- The curator approves a preferred schedule; the system changes the status to Approved by Curator and sends the visitor the corresponding update.
- The curator manually submits the approved visit information through the external USC campus-entry process; the system allows the status to be changed to Submitted for Campus Entry.
- The request is declined, cancelled, or completed; the system records the decision and history.

4.9.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.9-01	The public website shall allow a visitor to submit a formal museum visit request without creating an account.	High
REQ-4.9-02	The system shall require one Visit Request to contain at least one preferred date and time option and shall allow multiple preferred options.	High
REQ-4.9-03	The system shall store each preferred schedule as a separate child record linked to the visit request.	High
REQ-4.9-04	The system shall collect approved contact, organization, visitor-count, purpose, and other information needed to process the request.	High
REQ-4.9-05	The system shall allow optional vehicle information to be recorded separately and linked to the visit request.	High
REQ-4.9-06	The form shall validate required fields, schedule values, visitor count, contact information, consent, and approved input limits.	High

REQ-4.9-07	The system shall notify the authorized curator through an in-system alert and email when a new Visit Request is submitted.	High
REQ-4.9-08	The system shall allow the curator to view, search, filter, and review Visit Requests and compose outbound email messages or status updates through BioSphere.	High
REQ-4.9-09	The system shall support the Visit Request statuses Pending, Approved by Curator, Submitted for Campus Entry, Declined, Cancelled, and Completed.	High
REQ-4.9-10	The system shall allow the curator to send an outbound request for additional information to the visitor while retaining the Visit Request's applicable workflow status and recording the curator's outbound message. Information subsequently received through an external email account may be manually recorded by the curator.	High
REQ-4.9-11	The system shall send curator-initiated outbound Visit Request messages and status updates through BioSphere to the visitor's email address and record the available delivery result.	High
REQ-4.9-12	The system shall present approved visitor and schedule information in a consolidated format for the curator's manual USC process.	High
REQ-4.9-13	The system shall not directly submit visitor information to USC ISMIS or approve campus entry unless a separate official integration is authorized.	High

REQ-4.9-14	The system shall record status history, acting curator, timestamps, outbound messages sent through BioSphere, and relevant curator-entered decision or follow-up notes. Incoming external email messages shall not be automatically imported.	High
REQ-4.9-15	The system shall restrict visit-request personal data to authorized internal access.	High
REQ-4.9.16	The system shall allow the curator to select the approved preferred schedule and preserve the other submitted schedule options as part of the request history.	Medium

4.10 Static Public Museum Website

4.10.1 Description and Priority

Description. Provides approved static museum information, Gallery content, visitor guidance, General Inquiry and Visit Request access, QR exhibit access, and the FAQ Assistant while preventing exposure of the complete internal specimen inventory and confidential operational data. The Home, Gallery, Visit, and About page content is maintained through authorized frontend source-code updates rather than through a curator-facing content-management interface.

Priority. High

4.10.2 Stimulus/Response Sequences

- A visitor opens the public website; the system displays the Home page and navigation to Gallery, Visit, and About.
- The visitor selects Gallery; the system displays only approved static museum and collection information.

- The visitor selects Plan Your Visit; the system opens the Visit page and provides access to the General Inquiry and Visit Request forms.
- The visitor selects About; the system displays the approved museum background, contact information, personnel information, and location details included in the deployed frontend.
- An approved change to the static Home, Gallery, Visit, or About content is required; an authorized developer or designated USC/DCISM maintainer updates the frontend source code and redeploys the website outside the curator interface.

4.10.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.10-01	The public website shall provide Home, Gallery, Visit, and About areas.	High
REQ-4.10-02	The public website shall provide a Plan Your Visit action that opens the Visit area.	High
REQ-4.10-03	The public website shall not include an Events page in the MVP.	High
REQ-4.10-04	The Gallery shall display selected curator-approved museum and collection content without exposing the complete internal inventory.	High
REQ-4.10-05	The Visit area shall provide visitor guidance, inquiry access, and formal visit-request access.	High
REQ-4.10-06	The About area shall provide approved museum background, mission, authorized museum personnel information, contact details, and location information.	High

REQ-4.10-07	The Home, Gallery, Visit, and About page content shall be implemented as static frontend content and shall not be editable through the curator or restricted Developer interfaces.	High
REQ-4.10-08	The public website shall not expose internal storage locations, internal remarks, audit data, account data, or private visitor information.	High
REQ-4.10-09	The public website shall use responsive layouts suitable for supported desktop, tablet, and mobile screen sizes.	High
REQ-4.10-10	Approved changes to the static Home, Gallery, Visit, or About page content shall require a frontend source-code update and redeployment by authorized technical personnel according to the approved maintenance and turnover procedure.	High

4.11 Controlled Knowledge-Based FAQ Assistant

4.11.1 Description and Priority

Description. Provides a public rule-based FAQ Assistant that matches museum-related questions against active curator-approved knowledge and directs unanswered questions to the General Inquiry form.

Priority. High

4.11.2 Stimulus/Response Sequences

- A visitor selects a suggested question or enters a museum-related question; the assistant searches for active curator-approved FAQ knowledge.

- The visitor's question matches an active approved FAQ entry according to the configured matching rules; the assistant displays the curator-maintained answer and may show related approved questions.
- The question does not match an active approved FAQ entry; the assistant states that no approved answer is available and directs the visitor to the General Inquiry form.
- The curator creates, edits, activates, deactivates, or archives FAQ knowledge; the assistant uses only active approved entries.
- A visitor asks the assistant to submit, approve, cancel, or modify a Visit Request; the assistant does not perform the transaction and directs the visitor to the appropriate Visit Request form or curator communication channel.

4.11.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.11-01	The public website shall provide a controlled knowledge-based FAQ assistant for museum-related questions.	High
REQ-4.11-02	The system shall allow the curator to create, edit, activate, deactivate, and archive approved FAQ questions and answers.	High
REQ-4.11-03	The system shall support curator-approved alternative wording, keywords, and categories for matching.	High
REQ-4.11-04	The assistant shall answer only from active curator-approved FAQ knowledge.	High
REQ-4.11-05	The assistant shall not use unrestricted general knowledge, identify specimens or species, invent museum information, generate scientific	High

	conclusions, or submit, approve, modify, cancel, or otherwise process Visit Requests.	
REQ-4.11-06	When the visitor's question does not match an active approved FAQ entry according to the configured matching rules, the assistant shall state that no approved answer is available and direct the visitor to the General Inquiry form.	High
REQ-4.11-07	The FAQ Assistant shall not request or store personal information during the FAQ interaction and shall not display private visitor, specimen, storage-location, account, or operational data.	High
REQ-4.11-08	The assistant shall require an internet connection in the MVP.	High
REQ-4.11-09	The system shall record FAQ knowledge changes in the audit log.	High
REQ-4.11-10	The assistant interface shall clearly indicate that its answers are limited to curator-approved museum information.	High

4.12 QR Exhibit Management and Public QR Pages

4.12.1 Description and Priority

Description. Creates controlled unlisted exhibit pages and printable QR codes for selected Cataloged specimen records that are approved for public display.

Priority. High

4.12.2 Stimulus/Response Sequences

- The curator selects a Cataloged record approved for public display and creates an exhibit page.

- The curator chooses approved fields and media, publishes the page, and downloads or prints its QR code.
- A visitor scans the code; the system opens the correct published exhibit page.
- The curator unpublishes, disables, or archives the exhibit page; the existing public URL displays an appropriate unavailable state and does not expose previously published content.
- The curator edits approved content; the QR URL remains stable unless the page is intentionally replaced.
- The curator attempts to create or publish an exhibit page from an Uncataloged, Archived, or non-public-approved specimen record; the system rejects the operation and identifies the unmet requirement.

4.12.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.12-01	The system shall allow the curator to create, edit, publish, unpublish, disable, and archive a QR exhibit page for a Cataloged specimen record marked for public display or another curator-approved exhibit.	High
REQ-4.12-02	The system shall require public-display approval before a specimen can be used as a QR exhibit source.	High
REQ-4.12-03	The system shall allow the curator to control which approved fields, descriptions, images, and other public content appear.	High
REQ-4.12-04	The system shall assign each published QR exhibit page a unique public URL.	High

REQ-4.12-05	The system shall generate a printable QR code that opens the correct published page.	High
REQ-4.12-06	The system shall allow the curator to download or print an approved exhibit label containing the QR code and a human-readable direct URL to the same published exhibit page.	High
REQ-4.12-07	QR exhibit pages shall remain outside normal public website navigation and shall be accessed primarily through direct links or printed QR codes.	High
REQ-4.12-08	The system shall prevent storage locations, condition notes, internal remarks, audit data, and other restricted fields from appearing on QR pages.	High
REQ-4.12-09	A disabled or unpublished QR URL shall show a safe unavailable message and shall not expose previous private or unpublished content.	High
REQ-4.12-10	The system shall preserve the QR exhibit page's public URL when its approved content is updated, unless the curator intentionally replaces the page.	Medium
REQ-4.12-11	The system shall record QR publication changes in the audit log.	High
REQ-4.12-12	A visitor shall be able to access a published QR exhibit page through its direct public URL without requiring QR-scanning functionality on the visitor's device.	High

4.13 Selected WebAR Experiences

4.13.1 Description and Priority

Description. Adds optional browser-based augmented reality and three-dimensional viewing to a limited number of curator-selected and approved QR exhibits while keeping the normal exhibit information available independently of AR. Selection of specimens for AR presentation remains under curator discretion and may be guided by the specimen's relevance to a particular research or educational objective, conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value. These factors serve as selection guidelines rather than a fixed numerical scoring or automated ranking system.

Priority. High for fallback availability, security, and access control; Medium for the number of exhibits provided with AR experiences.

4.13.2 Stimulus/Response Sequences

- A visitor opens a published QR exhibit page with an active approved AR asset and selects **View in AR**; the system checks device compatibility, requests camera permission when required, and launches the supported experience.
- The visitor denies camera permission, uses an unsupported device, or the asset fails; the system keeps the normal exhibit page available and displays a fallback message.
- The curator selects and approves a specimen or exhibit for AR presentation based on the intended research or educational objective and applicable curator-determined considerations, which may include conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value. The Developer then deploys the approved AR asset, links it to the approved exhibit, and activates it after successful technical validation.
- The asset is disabled or removed; the system hides the AR action while retaining the normal QR page.

4.13.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.13-01	The system shall provide WebAR or 3D viewing only for selected curator-approved exhibits with available approved assets.	High
REQ-4.13-02	The system shall allow the curator to select and mark an approved exhibit for AR preparation through the authorized workflow. AR selection shall remain under curator discretion and may consider the specimen's relevance to a research or educational objective, conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value.	High
REQ-4.13-03	The developer shall be able to upload, update, activate, deactivate, or replace the AR asset linked to the approved exhibit.	High
REQ-4.13-04	A published exhibit page shall provide the View in AR action only when an approved usable AR asset is active and the required AR capability is available on the visitor's device and browser. When AR capability is unavailable, the system shall retain the regular exhibit content and shall not make AR necessary for accessing approved specimen information.	High
REQ-4.13-05	The system shall request camera permission only after the visitor explicitly starts the AR experience.	High

REQ-4.13-06	The normal QR exhibit page shall remain available when AR is unavailable, unsupported, denied, disabled, or fails to load.	High
REQ-4.13-07	AR content shall use only approved public exhibit information and shall not expose internal inventory or storage data.	High
REQ-4.13-08	Custom three-dimensional model production shall not be required for every specimen and shall depend on asset availability, documented legal authorization for use, budget, and development schedule. The system shall not automatically calculate, score, rank, or determine a specimen's eligibility for AR presentation based on the AR selection guidelines. Final AR selection and approval shall remain the responsibility of the authorized curator.	Medium
REQ-4.13-09	The system shall display an understandable message when an AR experience is unsupported, unavailable, or fails to load.	Medium

4.14 Selected Offline Inventory Operations and Synchronization

4.14.1 Description and Priority

Description. Allows an authorized curator to view and search previously synchronized specimen records and create new Uncataloged text-only drafts during temporary loss of internet connectivity. Offline-created drafts are stored locally and submitted to the server after connectivity is restored. Existing server records remain read-only while offline.

Priority. High

4.14.2 Stimulus/Response Sequences

- The curator signs in and synchronizes while online; the system stores the approved specimen data required for offline viewing.
- Connectivity is lost; the system displays an Offline state and allows the curator to view, search, and filter cached specimen records.
- The curator creates a new specimen record while offline; the system stores it as an Uncataloged local draft and marks it Pending Sync.
- Connectivity returns; after confirming that the curator remains authenticated, the system submits the pending draft to the backend.
- The server accepts the draft; the system marks it Synchronized and refreshes the local cached record.
- The server rejects the draft because of invalid, duplicate, or incomplete information; the system retains the local draft, marks it Sync Failed, and displays an understandable reason.
- The curator attempts an unsupported offline operation; the system blocks the action and explains that internet connectivity is required.

4.14.3 Functional requirements

ID	Requirement	Priority
REQ-4.14-01	The system shall allow the curator to view previously synchronized specimen records while temporarily offline.	High
REQ-4.14-02	The system shall allow searching and filtering of locally cached specimen records while offline.	High
REQ-4.14-03	The system shall allow the curator to create a new Uncataloged text-only specimen draft while offline.	High

REQ-4.14-04	The system shall mark every offline-created draft as Pending Sync until it is successfully accepted by the server.	High
REQ-4.14-05	The system shall retain an offline-created draft until it is synchronized or intentionally discarded by the curator.	High
REQ-4.14-06	The system shall submit pending offline-created drafts after stable connectivity is restored and the curator is authenticated.	High
REQ-4.14-07	The system shall retain a draft and display an understandable failure state when synchronization fails or the server rejects the draft.	High
REQ-4.14-08	The system shall prevent retry operations from creating multiple server records from the same offline draft.	High
REQ-4.14-09	The system shall display Online, Offline, Pending Sync, Synchronizing, Synchronized, and Sync Failed states.	High
REQ-4.14-10	Existing specimen records shall remain read-only while the application is offline.	High
REQ-4.14-11	Internet connectivity shall be required for editing existing records, cataloging, location changes, quantity or condition changes, images, imports, archiving, reports, inquiries, visits, accounts,	High

	notifications, publishing, backup, QR, and AR operations.	
REQ-4.14-12	Offline data stored in the browser shall not be treated as a server backup or confirmed server record until synchronization succeeds.	High

4.14.4 Offline Scope Summary

Scope	Behavior
Supported offline	View cached specimen records; search and filter cached records; create new Uncataloged text-only drafts; review Pending Sync and Sync Failed drafts.
Requires connectivity	Editing existing records; cataloging; location, lot, quantity, and condition changes; image uploads; imports; archiving; reports; General Inquiries; Visit Requests; accounts; email; public and FAQ content; backup and recovery; QR publishing; WebAR and 3D deployment.
Synchronization policy	Only newly created Uncataloged drafts are submitted. Existing server records are not edited offline, so online-versus-offline record conflict resolution is outside the MVP.
Device limitation	Cached records and pending drafts are stored within the authorized browser and device and may not be available from another device.

4.15 Audit Logging, Backup, and Recovery

4.15.1 Description and Priority

Description. Preserves accountability through protected audit records and provides approved backup, verified restoration, and turnover procedures for system data, configuration, and authorized assets.

Priority. High

4.15.2 Stimulus/Response Sequences

- A major internal action occurs; the system records the acting account, action, affected record/module, timestamp, and relevant result.
- A scheduled or authorized manual backup completes or fails; the system records the result and notifies the designated backup operator or hosting administrator when required.
- Authorized technical personnel initiate a restoration from a verified backup; the approved recovery procedure restores required data and records the event.
- Before turnover, the team performs and documents at least one complete backup-and-restoration test.
- An unauthorized user attempts to modify protected audit entries, initiate a backup, or perform a restoration; the system denies the operation and records the unauthorized attempt according to the security-logging policy.

4.15.3 Functional Requirements

ID	Requirement Description	Priority
REQ-4.15-01	The system shall record major internal actions including authentication events, curator account actions, specimen creation, editing, importing and archiving, lot splits or merges, quantity changes, location movements, inquiry and Visit Request status changes, FAQ-knowledge changes, QR-exhibit publication changes, AR deployment, report generation, backup results, and restoration activities.	High

REQ-4.15-02	Each audit entry shall contain the acting account, action, affected record or module, date/time, and result or relevant metadata.	High
REQ-4.15-03	The system shall prevent all users from editing or deleting protected audit entries through routine application functions.	High
REQ-4.15-04	Authorized users shall be able to search or filter audit information needed for review without exposing it publicly.	High
REQ-4.15-05	The deployment shall support scheduled backup of the PostgreSQL database, audit data, required object-storage assets, approved AR assets, and necessary system configuration.	High
REQ-4.15-06	The deployment shall record the result of every scheduled or authorized manual backup and shall notify the designated backup operator or hosting administrator when a backup fails.	High
REQ-4.15-07	Backup copies shall be protected through approved access controls and encryption appropriate to the deployment.	High
REQ-4.15-08	Authorized technical personnel shall be able to restore required BioSphere data from a valid verified backup using the approved recovery procedure.	High
REQ-4.15-09	Restoration actions shall be logged and shall not be presented as successful when verification fails.	High

REQ-4.15-10	Backup and recovery maintenance shall not require a general-purpose administrator dashboard in the curator application.	High
REQ-4.15-11	The development team shall complete and document at least one successful full backup-and-restoration test before formal turnover.	High
REQ-4.15-12	The final backup schedule, retention, recovery point, off-site destination, and responsible operator shall be approved by USC/DCISM and documented before deployment.	High

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-PERF-01	Under the controlled test environment and normal connectivity, at least 95% of standard page loads, searches, record views, and non-media record-saving operations shall complete within three seconds, excluding report generation, synchronization, and third-party notification delivery.
NFR-PERF-02	Indexed and paginated searching or filtering of a test dataset containing up to 60,000 specimen records shall return the first result page within three seconds under the approved test environment.
NFR-PERF-03	Monthly consolidated reports shall complete within 15 seconds and yearly consolidated reports within 30 seconds under the approved test environment.

NFR-PERF-04	Synchronization of 100 valid offline-created Uncataloged text-only drafts shall complete within two minutes after stable connectivity is restored and the curator is authenticated, under the approved controlled test environment.
NFR-PERF-05	The system shall support at least three concurrent authenticated users and 25 concurrent public visitors under the controlled test environment without exceeding the applicable response-time thresholds.
NFR-PERF-06	The interface shall show progress for operations expected to exceed three seconds and shall prevent duplicate submissions while processing.

5.2 Safety Requirements

ID	Requirement
NFR-SAFE-01	The system shall require confirmation before archiving, deactivating, removing access, discarding pending offline changes, or performing other high-impact operations.
NFR-SAFE-02	Routine curator operations shall archive official specimen records rather than permanently delete them.
NFR-SAFE-03	The system shall not silently overwrite conflicting synchronized data.
NFR-SAFE-04	The system shall not report a failed save, import, synchronization, backup, notification, or restoration as successful.
NFR-SAFE-05	The deployment shall maintain verified recoverable backups according to the approved backup policy.

NFR-SAFE-06	AR or camera features shall request camera permission only after a user-initiated action, and the normal QR exhibit page shall remain available as a non-camera fallback.
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5.3 Security Requirements

ID	Requirement
NFR-SEC-01	The system shall protect curator and developer functions through authenticated access.
NFR-SEC-02	The system shall enforce least privilege and backend authorization for every protected operation.
NFR-SEC-03	Passwords shall be protected through an approved salted password-hashing method or approved managed authentication provider and shall not be stored as plain text.
NFR-SEC-04	Production communication shall use HTTPS with a valid TLS configuration.
NFR-SEC-05	Authenticated sessions shall use an approved secure session mechanism, enforce session expiration, and require server-side validation for every protected request.
NFR-SEC-06	The system shall validate and sanitize submitted fields, query parameters, imported rows, and uploaded files before processing or storage.
NFR-SEC-07	Uploaded specimen images and three-dimensional or AR assets shall be validated against approved file types, file-size limits, required metadata, technical compatibility, and access rules.

NFR-SEC-08	Public interfaces shall not expose internal specimen fields, storage data, visitor data, accounts, audit logs, or private media.
NFR-SEC-09	The system shall collect only visitor information required for inquiry and visit processing and shall present the approved privacy notice and consent wording.
NFR-SEC-10	Private General Inquiry and Visit Request data shall be accessible only to authorized curator accounts and shall not be available through the routine Developer interface.
NFR-SEC-11	Audit entries and backup copies shall be protected from unauthorized alteration or access.
NFR-SEC-12	Secrets, service credentials, and environment configuration shall not be committed to the public source repository or exposed to client-side code.
NFR-SEC-13	The system shall apply database constraints, unique identifiers, and authorization checks to preserve official data integrity.
NFR-SEC-14	The project shall perform and document security testing for authentication, session handling, authorization, account-role separation, input and file validation, public/private data separation, and secure transport before turnover.
NFR-SEC-15	The system shall apply rate limiting or equivalent abuse controls to authentication, account-recovery, public forms, and FAQ Assistant requests.
NFR-SEC-16	Password-reset or account-recovery links and tokens shall be time-limited, single-use, securely generated, and invalidated after successful use.

5.4 Software Quality Attributes

ID	Attribute	Requirement
NFR-USA-01	Usability	The system shall use clear labels, validation messages, confirmation prompts, search controls, status indicators, and consistent terminology.
NFR-USA-02	Responsive Design	Public and authenticated interfaces shall adapt to supported desktop, tablet, and mobile screen sizes.
NFR-ACC-01	Accessibility	Public and authenticated user interfaces shall support keyboard navigation, visible focus indicators, meaningful form labels, sufficient contrast, alternative text for informative images, and understandable error messages.
NFR-REL-01	Reliability	Pending offline changes shall remain available until synchronized, resolved, or intentionally discarded.
NFR-REL-02	Failure Handling	The system shall display understandable errors and preserve recoverable user input where safe and practical.
NFR-REC-01	Recoverability	Required system data shall be restorable from a verified backup using documented procedures.
NFR-COMP-01	Browser Compatibility	The system shall support current versions of commonly used modern browsers approved during testing.

NFR-COMP-02	Information Accessibility	QR exhibit information shall remain accessible without AR support, and published exhibit labels shall provide a direct-URL fallback when QR scanning is unavailable.
NFR-COMP-03	AR compatibility	The system shall detect or appropriately handle unsupported AR environments and shall provide a non-AR fallback without blocking access to the regular exhibit page.
NFR-MNT-01	Maintainability	The source code shall use a modular structure, version control, documented configuration, migrations, and traceable requirement identifiers.
NFR-MNT-02	Turnover Documentation	The project shall provide deployment, backup, recovery, account-provisioning, user guidance, API/configuration, and basic maintenance documentation.
NFR-PORT-01	Portability	Docker and Docker Compose, together with documented environment variables and deployment procedures, shall support deployment to the approved USC/DCISM Linux environment or equivalent authorized infrastructure.
NFR-TEST-01	Testability	Critical business rules and API operations shall be structured so they can be verified through automated and manual tests.
NFR-DATA-01	Data Integrity	The system shall use validation, transactions where appropriate, database

		constraints, and audit history to reduce inconsistent official records.
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5.5 Business Rules

Code	Name	Rule
BR-01	Unique Identifier	Every specimen record shall have a unique system-generated specimen ID. The accession number shall remain nullable until assigned. Its official format, uniqueness scope, and assignment procedure remain subject to final curator validation, after which the approved rule shall be enforced.
BR-02	Uncataloged Rule	A newly saved record missing any applicable required non-image field shall be Uncataloged.
BR-03	Cataloged Rule	A record shall become Cataloged only when all applicable required non-image fields contain valid data or an approved Unknown/Not Applicable value.
BR-04	Archive Rule	Inactive official specimen records shall be archived rather than permanently deleted through routine curator operations.
BR-05	Specimen-Lot Grouping	One specimen record may contain one or more lots; each lot has one location, one condition class, and one positive quantity.
BR-06	Partial Group Change	A partial location or condition change shall reduce the source lot and create or update the target lot by the same quantity.

BR-07	Total Preservation	Moving or changing the condition of a lot shall not change the total quantity of the parent specimen record.
BR-08	Authorized Quantity Change	The total quantity shall change only through an authorized quantity adjustment, such as an addition, removal, transfer out, loss, destruction, or data correction.
BR-09	Separate Records	The same species may have separate records when physical grouping, collector, donor, storage assignment, or other curator-approved distinctions differ.
BR-10	Public Approval	Only curator-approved information may appear on the static public website, Gallery, FAQ Assistant, QR exhibit pages, or AR experiences. Approval does not require the curator to directly edit the static website; approved Home, Gallery, Visit, and About changes shall be implemented by authorized technical personnel.
BR-11	Storage Confidentiality	Detailed internal storage locations shall remain restricted to authenticated internal inventory functions.
BR-12	Developer Restriction	The developer is a restricted technical actor and shall not be treated as a curator or routine museum operator.
BR-13	No Public Registration	Public users shall not register for curator or developer accounts.
BR-14	Email Scope	Operational email alerts to the curator shall be limited to newly submitted General Inquiries and

		Visit Requests. Other authorized email uses are visitor inquiry and Visit Request communications, secure curator onboarding, password recovery, and optional password-change confirmation. Other system events shall use in-system alerts or audit records.
BR-15	External Approval	USC campus-entry processing and approval shall remain outside BioSphere unless a separately authorized integration is approved.
BR-16	Offline Position	Offline support is a secondary continuity capability limited to read-only access to cached specimen records and creation of new Uncataloged text-only drafts. Existing records shall not be edited offline, and offline data shall not replace server backup.
BR-17	AR Limitation	AR shall be limited to selected approved exhibits and shall not be required for every specimen.
BR-18	FAQ Boundary	The FAQ assistant shall answer only from active curator-approved knowledge and shall direct unmatched questions to General Inquiry.
BR-19	Gallery Boundary	The public Gallery is curated educational content and shall not become a complete searchable public inventory.
BR-20	Publication State	Uncataloged or Archived specimen records shall not be eligible for public exhibit publication.

		Unpublished, disabled, or archived public pages shall not expose previously published content.
BR-21	Outbound Email Boundary	BioSphere may send authorized email messages but shall not access or synchronize Gmail, Outlook, or another external mailbox. Only outbound messages sent through BioSphere and their available delivery results are automatically recorded. Relevant information received outside BioSphere may be manually recorded by an authorized curator.
BR-22	AR Selection Authority	Specimens for AR presentation shall be selected under curator discretion based on their relevance to a particular research or educational objective and may consider conservation status, rarity, endemism, exotic or invasive status, and economic or ecological value. These considerations are selection guidelines rather than a fixed scoring system, and BioSphere shall not automatically rank specimens or determine AR eligibility.

6. Other Requirements

6.1 Data, Privacy, and Retention

- The project shall prepare and obtain approval for visitor privacy notices and consent wording before pilot use.
- The final visitor-data retention period, deletion or anonymization procedure, authorized access, and response to privacy requests shall be approved by USC-BM and the appropriate university office.

- Only the minimum data required to process inquiries and visits shall be collected.
- Private visitor data, internal locations, account data, and audit information shall be excluded from public reports and pages.
- The deployment shall document where database records, object-storage files, logs, and backups are stored and who is responsible for them.
- Data used for testing shall be synthetic, anonymized, or formally authorized. Real specimen pilot records may be used with curator approval, while real visitor information shall require separate authorization and appropriate privacy safeguards.
- Approved retention and deletion procedures shall apply to active records, exported files, logs, and backup copies where technically and legally applicable.

6.2 Deployment, Backup, Maintenance, and Turnover

- BioSphere shall be deployable in a university-approved environment, preferably infrastructure designated or managed by DCISM, subject to assessment and authorization.
- The final deployment shall document frontend, API, database, object storage, domain/TLS, email, email-service configuration, backup destination, environment variables, scheduled jobs, and monitoring responsibilities.
- The final backup frequency, retention periods, recovery-point target, recovery-time target, encryption requirements, and off-site backup arrangement shall be determined with USC/DCISM based on available infrastructure, data-change frequency, storage capacity, and university policy. The approved values shall be documented before pilot deployment and turnover.
- The team shall provide source code, database migrations, deployment steps, backup/restore steps, account-provisioning steps, user guidance, system limitations, and known issues at turnover.

- USC-BM and USC/DCISM shall identify the post-turnover system owner, authorized curator account holder, backup operator, and technical maintainer.
- Formal turnover shall include verification of deployment access, designated account ownership, backup availability, restoration procedures, documentation completeness, and acknowledgement by the authorized client representative.
- The student team shall not be responsible for indefinite maintenance after formal turnover. Any agreed post-turnover support or defect-correction period shall be defined and documented separately.

6.3 Pilot Dataset and Data Migration

- The pilot deployment shall use approximately 500 curator-provided and curator-validated specimen records from the main-gallery collection. The final number may vary slightly based on record availability and validation results.
- The pilot shall not require complete digitization of the museum's entire collection, including the substantially larger insect and beetle holdings.
- Spreadsheet and CSV migration shall use an approved BioSphere import template with documented column mappings and validation results.
- The curator or assigned museum representative shall decide how incomplete, conflicting, or possible duplicate source records are corrected.
- Migration acceptance criteria shall include valid field values, correct Cataloged or Uncataloged status assignment, preserved source references where applicable, accurate specimen-lot quantities, valid storage assignments, and a documented error list for rejected or unresolved rows.

6.4 Scope Limitations and Non-Goals

Limitation / Non-Goal	Description
No OJT module	Student OJT tracking and management are outside the MVP.
No direct ISMIS integration	BioSphere does not submit to or replace the official USC campus-entry process without separate authorization.
No automatic campus approval	Only authorized USC personnel can approve campus entry.
No full public inventory	Public pages expose only curator-approved content and no searchable complete specimen catalog.
Limited AR	AR covers selected exhibits with approved compatible assets; no AR for every specimen.
No mandatory custom 3D models	Custom three-dimensional model production depends on asset availability, documented legal authorization for use, cost, team capability, and development schedule.
No native mobile application	The solution is delivered as responsive websites and a PWA.
Inventory-focused offline mode	Public forms, notifications, publishing, backups, current requests, and AR loading require connectivity.
Single museum unit	The MVP is for the USC Biological Museum only.
No general-purpose administrator dashboard	Developer and technical functions remain minimal and restricted.

No unrestricted chatbot	The FAQ assistant cannot answer outside curator-approved museum knowledge.
No perpetual student support	After formal turnover, maintenance responsibility transfers to the designated USC-BM or USC/DCISM personnel according to the approved turnover agreement.
No SMS integration	The MVP does not include an SMS gateway, SMS alerts, SMS replies, SMS credits, telephone-number configuration for notifications, or SMS-provider maintenance.
No general operational email alerting	Email is not used for synchronization failures, incomplete records, backup events, reports, specimen or location changes, QR events, AR events, or other ordinary internal system activity. These events use in-system alerts or audit records.
No offline editing of existing records	Existing specimen, location, lot, quantity, condition, media, and publication records are read-only while offline.
No general public website CMS	The MVP will not provide curator-facing forms or dashboard controls for editing the Home, Gallery, Visit, or About pages. These pages will use approved static frontend content. Changes require an authorized source-code update and redeployment by the designated developer or USC/DCISM maintainer. FAQ knowledge and QR exhibit pages remain curator-managed within BioSphere.

No offline conflict-resolution engine	The MVP does not compare or merge competing online and offline versions because existing records cannot be edited offline.
Device-local offline data	Cached records and unsynchronized drafts are stored in the current authorized browser and are not guaranteed to appear on another device.
No external mailbox integration	BioSphere will not connect to Gmail, Outlook, or another mailbox to retrieve incoming messages, detect visitor replies, import email threads, synchronize inbox content, or automatically attach external replies to General Inquiries or Visit Requests.

6.5 Verification and Acceptance Approach

Requirement Area	Primary Verification
Authentication and roles	Authentication tests, role-based authorization tests, negative-access tests, curator-onboarding tests, password-change tests, expired or reused onboarding-link tests, session-expiration tests, and password-recovery tests.
Specimen records	Tests using complete, incomplete, invalid, archived, imported, possible duplicate, and public-approved records.
Specimen lots	Quantity split/merge tests, partial move tests, partial condition tests, total-preservation tests, and invalid-quantity tests.

Storage hierarchy	Parent-child, leaf assignment, container movement, ancestor inference, privacy boundary, and location-search tests.
Search and reports	Search and report accuracy tests using seeded data; output-format validation; confidentiality checks; and response-time tests using the approximately 500-record pilot dataset and a generated larger performance dataset where required.
Offline behavior	Disconnect and reconnect tests; cached viewing tests; offline search and filtering tests; offline Uncataloged-draft creation tests; successful synchronization tests; failed-synchronization preservation tests; duplicate-retry prevention tests; and verification that unsupported editing actions are blocked while offline.
Public website	Responsive UI tests, content publication tests, accessibility checks, navigation checks, and public/private data separation tests.
Inquiry and visits	Form-validation tests, workflow and status tests, multiple-schedule tests, in-system curator alert tests, outbound curator email-alert tests, outbound visitor message and status-update tests, delivery-result tests, verification that no external mailbox access or inbound-email import occurs, privacy tests, and curator User Acceptance Testing.
FAQ assistant	Approved match, alternative wording, inactive knowledge, no-match fallback, prohibited general answer, and inquiry redirection tests.

QR and AR	QR destination, printed direct-URL fallback, unpublished/disabled state, approved fields, representative device and browser compatibility, camera permission, supported AR launch, unsupported-device handling, AR-load failure, and regular exhibit-page fallback testing.
Security and privacy	Authentication, session handling, role-based authorization, rate limiting, input and file validation, secure transport, storage access, public/private data separation, consent, and data-minimization review.
Backup and recovery	Backup creation, protected storage, integrity verification, full restoration, and documented recovery test.
User acceptance	Curator User Acceptance Testing for authenticated museum-management functions, together with a target of approximately fifty (50) representative public participants for the public website, Gallery, FAQ Assistant, General Inquiry form, Visit Request form, QR exhibit pages, and selected WebAR experiences. Public participants may include actual museum visitors, USC students, faculty, staff, and other individuals who reasonably represent potential museum visitors. The final number shall depend on participant availability and the approved testing methodology.

Public-Facing System Evaluation Participants

Based on the curator interview documented in Appendix C, Question 9, educational tours may involve groups of approximately fifty (50) to one hundred (100) museum visitors. When an educational tour is available during the system

evaluation period, willing visitors may be invited to participate in evaluating the public-facing components of BioSphere.

The project shall target approximately fifty (50) representative public participants. If an educational tour is unavailable within the testing schedule, participants may include USC students, faculty members, staff members, and other individuals who reasonably represent potential museum visitors. The actual number and composition of participants shall be reported accurately in the final study.

6.6 Information Requiring Verification Before Version 1.0

Item	Verification Needed
Accession number	Official format, assignment rule, uniqueness handling, and treatment of records without an existing number.
Faculty/client sign-off	Approved requirement baseline, user roles, curator onboarding policy, acceptance criteria, and authorized representatives.
Deployment environment	DCISM Linux version, server resources, runtime, PostgreSQL arrangement, object storage, domain/TLS, outbound email, firewall/network rules, and permissions.
Reports and exports	Final five report layouts, fields, confidentiality rules, and approved output formats.
Visitor privacy	Approved privacy notice, consent wording, retention period, authorized access, deletion/anonymization, and breach/escalation contacts.
Notifications	Official sender email address, authorized curator recipients, official Reply-To museum email address, approved General Inquiry and Visit Request templates, visitor status-update templates, account-security

	templates, delivery-failure handling, and responsibility for maintaining the outbound email service after turnover. External mailbox access and incoming-email synchronization are excluded.
Data migration	Which files and paper records are included, who cleans them, mapping rules, pilot acceptance criteria, and unresolved-data handling.
QR labels	Label size, print format, physical placement, replacement procedure, and who owns printing costs.
AR assets	Final selected exhibits, available assets, documented legal authorization or licensing, device and browser targets, maximum file sizes, asset optimization requirements, and responsible asset creator or provider. The curator-approved AR selection guidelines have been confirmed; however, the final specimens to receive AR content remain subject to asset availability, technical compatibility, development feasibility, and curator approval.
Backup policy	Final schedule, retention, off-site destination, encryption, recovery objectives, operator, and testing frequency.
Internal mockups	Latest approved screenshots or HTML mockups for the Dashboard, Specimens, Cataloging, Locations, Reports, Curator Account Management, FAQ Knowledge Management, Restricted Developer Interface, and AR Asset Deployment, when interface figures are required in the SRS.
Public UAT participants plan	Confirm with the faculty adviser and client the target of approximately fifty (50) representative public participants,

	the acceptable participant categories, the recruitment method, and the minimum feasible participant count if an organized museum visitor group is unavailable during the testing period.
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Appendix A: Glossary

Term	Definition
Accession Number	A museum-assigned identifier or code. Its official BioSphere format and assignment rule are still subject to curator validation.
Archived Record	An inactive specimen record retained for history and audit but excluded from normal active inventory and public eligibility.
AR / WebAR	An augmented-reality experience launched from a compatible web browser, normally using a device camera.
BioSphere	The proposed USC-BM inventory, visitor coordination, public website, QR/AR exhibit, FAQ, audit, and backup system.
Cataloged	A specimen status assigned after all applicable required non-image fields are valid or use approved Unknown/Not Applicable values.
Controlled Knowledge-Based FAQ Assistant	A public rule-based assistant that answers only from active curator-approved museum FAQ knowledge.
Curator	The authenticated operational museum user responsible for records, locations, requests, reports,

	public content, FAQ knowledge, and authorized account handover.
Developer	A restricted technical actor responsible only for approved initial curator/access actions and AR asset deployment.
Gallery	The public website area containing selected approved educational museum and collection content; it is not the complete internal inventory.
General Inquiry	A public question or message that is not automatically treated as a formal visit request.
ISMIS	A USC system/process referenced for campus-entry handling; BioSphere has no approved direct integration in the MVP.
Limited Offline Inventory Support	A secondary BioSphere capability that allows authorized curators to view and search cached specimen records and create new Uncataloged text-only drafts during temporary connectivity interruptions. Existing records are not editable offline.
Pending Sync	A local change waiting to be sent to the server after connectivity and authentication are available.
Public Display Approval	A curator-controlled approval allowing selected data to be used as a source for public exhibit content; it does not automatically publish the full record.
QR Exhibit Page	An unlisted public page opened primarily through a printed QR code or direct link and containing only approved exhibit information.

Specimen Lot	A quantity subgroup of one specimen record with one storage location and one condition class.
Storage Unit	A node in the physical location hierarchy, such as room, cabinet, drawer, shelf, box, or vial.
Uncataloged	A saved specimen record that does not yet satisfy all applicable required non-image fields.
Visit Request	A formal public request containing visitor information and one or more preferred schedules, subject to curator and external USC processing.

Appendix B: Analysis Models

B.1 System Context Model

Actor	Interface	Interaction Boundary
Public Visitor	Public Website	Approved museum information, Gallery, Visit/About, inquiry, visit request, FAQ, QR, and selected AR.
Museum Curator	Curator PWA	Accounts/handover, inventory, specimen lots, locations, requests, inquiries, reports, website/FAQ/QR content, audit, backup controls, and offline work.
Developer	Restricted Developer Interface	Initial curator/access administration when authorized and approved AR asset deployment only.

USC/DCISM Technical Personnel	Deployment Environment	Server, database, storage, TLS, \ backup/recovery, and turnover maintenance.
USC Campus-Entry Process	External Manual Process	Receives approved information through curator action outside BioSphere; no direct integration in MVP.

B.2 Core Data Entity Relationships

Entity	Relationship Summary
UserAccount	One account belongs to one approved role; curator accounts are separate. Account actions create audit entries.
Specimen	One master record has one status and may have many SpecimenLots, images, public exhibit records, and audit entries.
SpecimenLot	Belongs to one Specimen; references one StorageUnit; stores one condition and positive quantity.
StorageUnit	Self-referencing parent-child hierarchy. A valid leaf or holds-specimens unit may contain many SpecimenLots.
VisitRequest	Has one requester, many PreferredVisitDate records, zero or more linked vehicle records, messages, and status-history entries.
GeneralInquiry	Has requester/contact data, message, status, curator replies, and optional referral to VisitRequest.

FAQKnowledge	Contains approved question, answer, alternatives, keywords, category, status, and audit history.
ExhibitPage	References approved specimen or exhibit content; has publication state, selected public fields, unique URL, QR asset, and optional AR asset link.
ARAsset	Linked to selected ExhibitPage; has file metadata, approval/deployment state, and developer audit history.
AuditLog	References acting account, action, module/record, timestamp, result, and relevant change metadata.
Notification	References recipient/actor and related inquiry, visit, sync, or operational event; may have email status.

B.3 State-Transition Models

Model	Allowed Transition Summary
Specimen	Uncataloged -> Cataloged when applicable required fields are complete; Uncataloged or Cataloged -> Archived when made inactive; Archived may be restored only through an approved curator action if implemented.
General Inquiry	Pending → Reviewed → Closed; Pending or Reviewed → Turned to Visit Request. All transitions shall be recorded.
Visit Request	Pending → Approved by Curator → Submitted for Campus Entry → Completed. A request may become Declined or Cancelled according to the approved workflow.
QR Exhibit Page	Draft/Unpublished -> Published -> Unpublished or Disabled; public access is safe only while Published and eligible.

Offline Change	Pending Sync → Synchronizing → Synchronized; or Pending Sync/Synchronizing → Sync Failed → Retry or Intentional Discard.
Curator Account	Invited/Created -> Active -> Inactive; handover transfers operational responsibility without sharing credentials.

B.4 Problem-Objective-Feature-Test Alignment

Problem	Objective	Feature	Testing Method	Success Metric
Fragmented paper/spreadsheets inventory	Centralize and protect specimen records.	Specimen Inventory, Import, Search, Audit, Backup.	Black-box record tests, import validation, search accuracy, backup/restore, curator UAT.	Accurate retrieval; validated imports; successful restoration; curator acceptance.
Difficult location and quantity tracking	Track physical storage and grouped quantities.	Storage Hierarchy and Specimen Lots.	Move, split, merge, ancestor, and total-preservation tests.	No quantity loss; valid locations; complete movement history.

Manual inquiries and visit coordination	Structure public communication and curator review.	Inquiry, Visit Request, Notifications, Reports.	Form/workflow/email tests and curator UAT.	Valid submissions, correct status transitions, recorded responses.
Limited approved public access	Provide controlled public information without exposing internal data.	Website, Gallery, FAQ, QR, AR.	Publication/privacy tests, FAQ match/fallback, QR/AR device tests.	Only approved fields visible; fallback works; no internal data exposure.
Temporary connectivity interruptions	Continue selected inventory work and synchronize safely.	PWA Cache, IndexedDB, Sync Queue, Conflict Review.	Cached viewing and searching, offline Uncataloged-draft creation, reconnection, draft synchronization, failed-sync preservation, duplicate-retry prevention, and verification that unsupported	Pending changes preserved; conflicts not overwritten; target sync time met.

			offline actions are blocked.	
Risk of record loss and uncertain turnover	Provide recoverability and maintainable handover.	Backups, Restore, Audit, Documentatio n, Docker deployment.	Full restoration test, documentation review, turnover checklist.	Verified restore and complete approved turnover package.